Rural-Urban Risk Sharing Networks: Uber Driver Labor Supply Responses To Shocks In Uganda*

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2/20/23

This paper examines the effect of flexible labor arrangements on insuring against rural weather shocks in developing countries, using data from the recently formalized taxi industry. I use machine learning classification to match Uber drivers in Kampala to rural weather events. A one standard deviation increase in the intensity of a shock is associated with working 20% more in the week of a shock and then hours decline steeply the following week. Dynamics are driven by extensive margin changes and are stronger for poorer parts of Uganda. Driver behavior changes based on the type of shock; drivers tend to drive more intensively and increase income if a drought event occurs, but leave the platform in the case of a flood.

JEL: O12, O13, O17, O18, J01, C01

Keywords: rideshare, Uganda, machine learning, taxi, informality, risk-sharing, migration, labor, drought, flood

*Thank you to my committee members, Dalia Ghanem, Travis Lybbert and my chair, Michael R. Carter for constantly pushing me to improve the paper. I thank Brian Dillon and Aditya Vashistha, my mentors at Cornell for helpful discussions. I like to thank NEUDC, PacDev, WGAPE and AAEA for giving me a platform to present my work and get comments. Thank you to Uber for providing the driver data and thoughts. Thank you to Samuel Bird providing the voter registration data and the chance to collaborate.

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1 Introduction

Rural-urban linkages are an important factor in poverty alleviation and economic development. Often a key driver of these linkages is remittances, which can act as insurance against rural weather shock risk (Kazianga and Wahhaj 2020; Stobl and Valfort 2013; Mueller and Osgood 2009; McKay and Deshingkar 2014; Saptono, Mahmud, and Lei 2022; Yang 2011) or as relief from wage rigidity (Kaur 2019) and volatility (Rosenzweig and Udry 2014).[1] These types of linkages set lower-middle income countries (LMICs) apart from more developed nations, in that rural-urban income streams are crucial for household livelihood.

This paper provides novel evidence for how urban migrants working in formerly informal employment, leverage their labor market flexibility and react to shocks in rural areas. Uber, the platform of focus, entered the market of a largely informal sector of the economy, and the availability of data makes it easier to observe a previously obscure sector (Lakemann and Lay 2019).

Determining flexibility in the labor market presents challenges, given current data limitations. Often the only data available are self-reported at yearly time scales/intervals. Moreover, existing labor supply data may not have information on rural linkages. I tackle these challenges by connecting rural weather shock data with high-frequency Uber data using machine learning techniques. I use the ethnolinguistic signal in a driver's surname to trace their rural area of origin, by using a third external dataset on voter registration, which has information on each Ugandan that registered to vote in 2016. The plausibility of using name data to signal

[1] These linkages are often buttressed by ethnic or clan relationships that persist through time (Gelan 2002).

rural linkages is supported by both anecdotal and anthropological evidence.[2]

This work also adds to the empirical question of short-term and long-term responses to shocks. A large literature on labor market response to unemployment shocks and natural disasters highlight long-term changes to household income and labor market participation (Martinoty 2022; Chaijaroen 2019; Giannelli and Canessa 2022). However, there is also evidence for households making mid-season adjustments in response to changes in expected household income over a shorter timespan. The advent of mobile money has made it increasingly easy to send and receive money quickly when households are in need, making short-term responses to shocks less costly (Jack and Suri 2014; Munyegera and Matsumoto 2016). The answer to this empirical question is increasingly important, especially in regards to remittances as a form of insurance.

I develop a model that relates urban labor supply to different types of migrant aid and use it to inform my empirical strategy. My results show that Uber drivers in Kampala react to adverse weather shocks in their rural home regions and use the platform's flexibility to adjust hours in the *same week* as a shock to biomass growth. A one standard deviation increase in the intensity of an adverse agricultural shock leads to an average treatment effect of 7 hours online in the week of the event but with -5 hours online one week after the event (a change of 22% and -16% over average hours, respectively). Earnings follow similar dynamics with an effect of around 25% in the week of a shock and -12% one week afterward. I find evidence that this pattern is due to changes in the extensive margin of hours supplied. Drivers are nearly

[2]The approach in this paper is similar in spirit to past approaches of using surnames to predict ethnicity (Cruz, Labonne, and Querubin 2017; Lowes et al. 2017). In contrast, I use machine learning methods which have several advantages, described below.

40% more likely to be online in the week of a shock. But one week after, they are almost 50% *less* likely to be working on the platform. These effects are not observed in the case of a positive shock to biomass growth suggesting that drivers react to adverse shocks, not to a rural shock's effects on urban demand. I find no heterogeneous differences across whether the shock occurs during the planting or harvest seasons, suggesting that driver reactions are due to changes during the growing season when adverse shocks are the most influential to changes in household income.

Drivers may be substituting away from working on the platform in favor of helping their families cope with shocks. I do not observe these mechanisms directly, but to explore their plausibility, I use the insights from my model and an alternative shock index that can better capture differences between droughts and floods. I find that droughts lead to increases on the intensive margin in the same week as the shock by about 7 hours online, but decreases on the extensive margin by 50%. Weekly earnings increase by 60% that week, suggesting that drivers re-optimize to work less but at higher-earnings times. In the case of a flood, however, intensive margin, extensive margin and weekly earnings decrease suggesting that drivers leave the platform. These results are strongly in line with the theoretical framework posed by Chakraborty and Pandey (2021), which posits that shocks that primarily affect income will lead to more migration and remittance behavior, while natural disasters such as floods, will lead to recalling migrants, as the demand for labor services outweighs migrant income potential.

My model predicts that the initial wealth of the household changes the labor supply of a migrant in response to a shock. Drivers from the poorest region of Uganda, Karamoja, where

households depend on unconditional help from family as a way to cope with shocks, tend to have even longer spells off the platform and are affected by the shock after drivers from other regions are no longer affected.

This paper makes four contributions to the literature. First, I add to the literature on how how a formerly difficult to observe informal sector is being transformed by new technology. The entrance of rideshare companies is a little-studied topic and has only recently become more prevalent in the development literature. This topic is increasingly policy-relevant, and as technology companies begin to expand their operations into LMICs, it is important to gain a clear understanding of the benefits, costs, and potential structural implications of such an expansion.

Second, my analysis sheds light on the effects of a sector of the labor market which is often difficult to study. Unemployment continues to be a large problem in many LMICs. It is commonplace to see disguised unemployment in the form of larger shares of the population being engaged in forced self-employment and entrepreneurship occupations (Breza, Kaur, and Shamdasani 2021). As more labor is pushed into the informal sector, its economic impact becomes more important, but the ability to study it becomes more difficult (Schneider, Buehn, and Montenegro 2010). These oftentimes, flexible work arrangements allow the worker to change their hours quickly or to leave work altogether for spells of time, but they come with drawbacks, such as less job security and a lack of any formalized benefits. The results from my paper show that informal laborers utilize flexible labor arrangements because it may aid them in buffering against adverse shocks in the absence of formal insurance products.

Third, the machine learning procedure that I utilize can be applied to many contexts where surnames signal ethnic or regional membership/ties, thus providing a way to connect seemingly disparate datasets. This is an especially vital consideration when studying developing countries, where data gaps are prevalent. I provide a novel way to predict membership based on geographic origins, when such data is not available, by merging anthropological and linguistic evidence with data science.

Fourth, I develop an econometric estimator that uses machine learning predictions to account for misclassification bias, contributing to the methods literature on using machine learning in causal estimation. This estimator is a novel way of using machine learning results for causal estimation and merges classic econometric techniques grounded in economic choice with new techniques prevalent in epidemiology and the medical sciences. In contrast to endogenously determined switching probabilities in the classic econometrics literature (Heckman 1979), the probabilities in my case are exogenous and derived through machine learning. These probabilities are akin to multi-valued propensity scores being estimated by machine learning (Imbens 2000; Westreich, Lessler, and Funk 2010; B. K. Lee, Lessler, and Stuart 2010; Goller et al. 2020).[3]

The paper is organized as follows: Section 2 outlines the conceptual framework and describes how shocks translate into changes in labor supply. Section 3 gives some background about Uganda's economy, as well as anthropological evidence to support my using surnames in prediction. Section 4 describes the three datasets I use in the paper and how I merge them.

[3]This novel switching regression technique is available as an open-source python package `ml-switching-reg` and simulations tests for the estimator can be found at amichuda.github.io/ml-switching-reg-sim, which serves as part of this paper's supplementary appendix.

Section 5 provides the empirical framework, the classification procedure, and how I develop the switching regression estimator. Section 6 shows my results. Section 7 discusses the results and frames them in terms of coping strategies of households in rural areas. Section 8 concludes.

2 Conceptual Framework

This section describes a conceptual framework and stylized model of how urban labor supply response can translate to rural household aid. I discuss three aspects. The first if the degree of market integration within the country. Higher price integration across regions in the country will lead to faster changes in urban prices in response to rural shocks. Any impact to urban drivers could then be attributed to both the effect of the shock itself and to changes in urban demand. I argue that price integration across regions in Uganda is heterogeneous and is less integrated, giving me a short window where any reaction by drivers can only be attributed to the shock and not to demand as well.

Second is the nature of the driver response. In the case of this paper, drivers can make changes on the intensive or extensive margins, which can be attributed to an increase in labor supply to send more income, or a decreases on the extensive margin if the effects of the rural shock is labor-requiring, translating to the marginal utility of the household to extra labor being higher than lost wages on the platform.

Third is response timing. Drivers can choose to react to expected changes in future income due to a shock or to realized changes once the shock has occurred. This can be attributed to

the severity of a shock, the prevailing wage and the ensuing labor-leisure tradeoff, and other factors.

2.1 Price Transmission in Uganda

Do rideshare drivers react to the realized shock through changes in demand for rideshare; or do they respond to the shock, itself, through a household shock that changes expectations about future household income? This will determine whether effects can be plausibly associated with coming from remittances or other help, rather than a shift in demand for ride-share.

The determining factor for this is the degree of market integration. Markets in low-income countries are often less integrated. Market imperfections, infrastructure challenges, or high transaction costs increase the time for a price change to transmit across regions (Abdulai 2000; Sexton, Kling, and Carman 1991). The less integrated markets are, the less correlated prices will be across those markets in the short term. Baquedano and Liefert (2014), for instance, finds that the long-run consumer price transmission rate of world prices are relatively slow and incomplete in developing countries. Among maize-producing countries, only three adjust to changes in world prices in less than a month (The Dominican Republic, Malawi and Peru).

For the purposes of this paper, it is not necessary to prove that regions of Uganda do not transmit prices, only that this transmission takes longer than the time horizon of the study. In terms of Uganda, there is limited evidence in the literature on the spatial transmission of

prices in Uganda. One might suggest that since the country is small, there is no price integration problem, as it won't take too long for demand and prices to be affected. However, there are three main factors that impede the integration of prices across rural and urban areas in Uganda. Firstly, there are spatial breaks across the country that have historically lessened price integration across regions with the central market of interest in Kampala. Using weekly price data, Rashid (2004) shows that, post-liberalization maize prices in Uganda in the early 1990s exhibited limited to no price integration between the Northern and Central parts of the country. This is confirmed in Boysen et al. (2009), where higher distance from Kampala leads to slower adaptation to price changes. Although this is related to my case, there is limited evidence as to how supply shocks transmit to the urban market. Secondly, dumping of agricultural goods from other countries, and food aid makes urban prices insensitive to changes in rural circumstances (Kaspersen and Føyn 2010). Thirdly, there are significant price differentials across the country due to limited infrastructure and transportation costs. Remoteness from the city, infrastructure capacity, among others, exacerbate price differentials (Shinyekwa, Ijjo, et al. 2016).

2.2 Nature of Driver Response

We can observe drivers reacting to shocks on either the intensive or extensive margin. A driver may drive more in response to a shock to make more income to help their relatives, or they may leave the platform altogether when their physical labor is needed. This decision depends on the migrant's consumption-leisure tradeoff; the driver trades off lost wages with the marginal utility to the rural household. There is a large literature on the migrant

remittances affecting household labor, especially child and female labor supply (Amuedo-Dorantes and Pozo 2006; Airola 2008; Acosta 2006; Nguyen and Purnamasari 2011). There is also a large literature on using remittances as insurance as described above. However, it still remains an empirical question as to the labor supply elasticity of *migrant* labor supply to their aid.[4]

For instance, flood shocks have been documented to increase the need for labor services (Lewin, Fisher, and Weber 2012). In another study, Chakraborty and Pandey (2021) explores this in the context of Kyrgyzstan and lays out a theoretical framework for the incentives for in- and out-migration. They find that droughts, which primarily reduce income lead to a decreased likelihood of recalling a migrant, while large destructive natural disasters such as earthquakes have the opposite effect due to an increased demand for labor services. This sets up a relationship between the type of shock and intensive or extensive margin changes. If there is a drought, the household might need help financially; this would include needing money to buy food that would have initially been produced from the plot, or to buffer against an unexpected income shortfall as a result of reduced yields. In the case of something that destroys infrastructure or housing, there would be a need for the physical help of the migrant, so we would expect to observe drivers being on the platform less.

[4]The assumption that labor supply would respond to immediate changes to remittance allocation hinges on the assumption that saving for the migrant is difficult and that the migrant's only response is to change their labor-leisure ratio. If a migrant decides to save during periods when there aren't shocks, then they can send large inflows of remittances when a shock does affect the household.

2.3 Response Timing

The timing of an effect is also important. As discussed above, one of the choices that migrants have is to send aid as insurance, choosing to send money or labor in response to shocks. Drivers can react to the true realized losses of shocks as the result of a drought or a flood, when their household is already in dire straits and in critical need or they can react to changes in *expected loss* in income as the result of the shock. For instance, in the case of natural disasters, remittances are sent in anticipation of shocks (Mohapatra, Joseph, and Ratha 2012; Adams Jr 2011). The transmission of this information can be relatively quick, given mobile phone penetration in the area. Afterwards, migrants can also decide to help their family as long-run effects begin to take hold.

There is evidence of mid-season adjustments occurring in response to shocks (Fafchamps 1993; Jayachandran 2006). This can apply to urban labor adjustments to rural shocks as well. In response to violence, Fernández, Ibáñez, and Peña (2011) find that Colombian farmers adjust labor supply and begin to work in the off-farm sector, but are constrained by low labor demand. Labor supply adjusts in response to *ex ante* changes in expected shocks, leading to higher participation in the off-farm labor market or the creation of informal businesses (Rose 2001; Mathenge and Tschirley 2015). This implies that some household members might become migrants or off-farm workers in anticipation of oncoming shocks and react very soon after its realization.

On the other hand, labor supply adjustment can also occur over longer periods of time. In Argentina, Martinoty (2022) find that shocks to household employment can lead to persis-

tent increases in labor force participation. Chaijaroen (2019) shows that although fishery households experienced a large decrease in income due to coral bleaching in Indonesia, labor market effects only occurred in the long term. In response to floods in Bangladesh, affected households were more likely to migrate out and remittance income accounted for 40% of the income loss from the floods (Giannelli and Canessa 2022).

2.4 A Stylized Model

To formalize intuition, I present a stylized model based on the work in Zhu (2019) and in the spirit of Chakraborty and Pandey (2021), which presents a two-period model of an inter-linked migrant that decides the optimal levels of remittances to send to a household experiencing a shock. Rather than modeling some fixed sum of wealth, however, I also model the labor-leisure decision to explore how observed labor hours translates to household aid. Each period represents a response timing decision. Does the migrant make immediate funds available or do they decide to provide funds later for recovery, or both?

The model depicts a household that lives for two periods, and a shock has already been realized that has reduced household income W. The household must make an investment and savings decision for the next period that will be an input into a production function. If that shock has damaged infrastructure, the effectiveness of how that investment translates to production may also be reduced.

The household solves the following maximization problem:

$$\max_{S,I} u(c_1) + \beta u(c_2) \tag{1}$$

$$s.t. \tag{2}$$

$$c_1 \leq \epsilon W - I + R_1 - S \tag{3}$$

$$c_2 \leq \delta S + \alpha f(L_r, I) + R_2 \tag{4}$$

where β is the discount rate, c_1 and c_2 are consumption at periods 1 and 2, and δ is the discount rate of savings. ϵ is a shock that has reduced wealth, and α is an infrastructure shock that can reduce the effectiveness of investment. This is similar to the approach in Hallegatte, Jooste, and Mcisaac (2022) who model the macroeconomic consequences of natural disasters. This can signal that some investment is being used in the reconstruction of capital, such as time spent on getting rid of excess water from land, the increased effort needed to work in more difficult conditions or the reduced capacity of transportation.

R_1, R_2 and L_r are all variables that are chosen by the migrant: the remittances chosen in periods 1 and 2, and the labor of the migrant in the case that they return. $u(\cdot)$ is a utility function with $u' > 0$ and $u" < 0$. $f(\cdot)$ is a standard production function with $f' > 0$ and $f" < 0$. Once investment and savings decisions have been made, let $V(R_1, R_2, L_r) \equiv u(c_1^*) + \beta u(c_2^*)$ be the value function of the household, with $\partial V/\partial R_1 > 0$, $\partial V/\partial R_2 > 0$, $\partial V/\partial L_r > 0$.

The migrant has three ways of helping their household: they can choose to provide remittances right at the realization of the shock in the first period, which can help buttress the

household's investment ability. This is especially important if ϵ is high and the household's ability to produce goods is at risk. If, however, the migrant would like to help directly with disaster relief once the shock has been has been realized, then they can choose R_2 as a strategy. Finally, the migrant can choose to provide labor to complement the investment in the second period. This model makes a simplifying assumption in that the migrant provides *all* labor to second-period production, necessitating that they always provide $L_r > 0$ in equilibrium. However, for this model, since we are looking at how a migrant's choices affect their urban labor hours, it will suffice to see if there is a large increase as the result of a shock, not necessarily that they would choose $L_r = 0$.

The migrant takes the value function of the household as given and chooses R_1, R_2 and L_r to maximize their own two-period utility. The migrant's problem is:

$$(1-\theta)log\left[m(c_1, l_1) + \beta_{mig} m(c_2, l_2)\right] + \theta log(V(R_1, R_2, L_r)) \tag{5}$$

$$s.t. \tag{6}$$

$$c_1 + wl_1 \leq w(\bar{L_1} - l_1) - R_1 \tag{7}$$

$$c_2 + wl_2 + wL_r \leq w(\bar{L_2} - l_2 - L_r) - R_2 \tag{8}$$

where m is a utility function satisfying the standard conditions, l is the amount of leisure time, β_{mig} is the migrant's discount rate, w is the prevailing urban wage and $\bar{L_{1,2}}$ is the migrant's labor allocation. We normalize the price of consumption goods to 1, and assume, without loss of generality, that w stays the same across both periods. θ is a preference

parameter for how much the migrant cares about the household, versus themselves. For simplicity, the model assumes that money can be transferred costlessly. Although there are fees associated with sending mobile money in developing countries, this does not change the results substantively. We also define the amount of labor that the migrant spends working in equilibrium as:

$$L_1^{p*}(\epsilon, \alpha, W) \equiv \bar{L}_1 - l_1^*(\epsilon, \alpha, W) \tag{9}$$

$$L_2^{p*}(\epsilon, \alpha, W) \equiv \bar{L}_2 - l_2^*(\epsilon, \alpha, W) - L_r^*(\epsilon, \alpha, W) \tag{10}$$

There are three main questions that this model can help elucidate:

1. What determines whether a migrant will send remittances or take part in labor?
2. How does the severity of a shock affect the migrant's equilibrium choices, and
3. How does initial household wealth play into these choices?

The migrant's problem cannot be solved analytically, so computational methods were employed to solve for the optimal solutions. In order to understand how the shock changes behavior, I also compute a baseline simulation, where I take the extreme case of $\theta = 0$. This allows me to see what labor supply would be like without any reaction to shocks.

To understand the mechanisms behind this labor supply effect, I plot out how initial wealth of the household plays into the migrant's choices, shown in Figure 1. Less poor household migrants tend to use L_r as a means of helping the household, apart from a small increase in

R_1 and R_2 at very low levels of ϵ. In this case, the household has enough resources to make enough of an investment into their farm in order to produce in the second period.

For poor households, depending on the severity of the shock, migrants use a combination of labor and remittances in order to help the household. At lower levels of ϵ migrants choose to have $R_1 = 0$, $R_2 > 0$ and then switch to a regime of $R_1 > 0$, $R_2 = 0$. There is as a trade-off between making early mid-season adjustments, and ex-post aid.

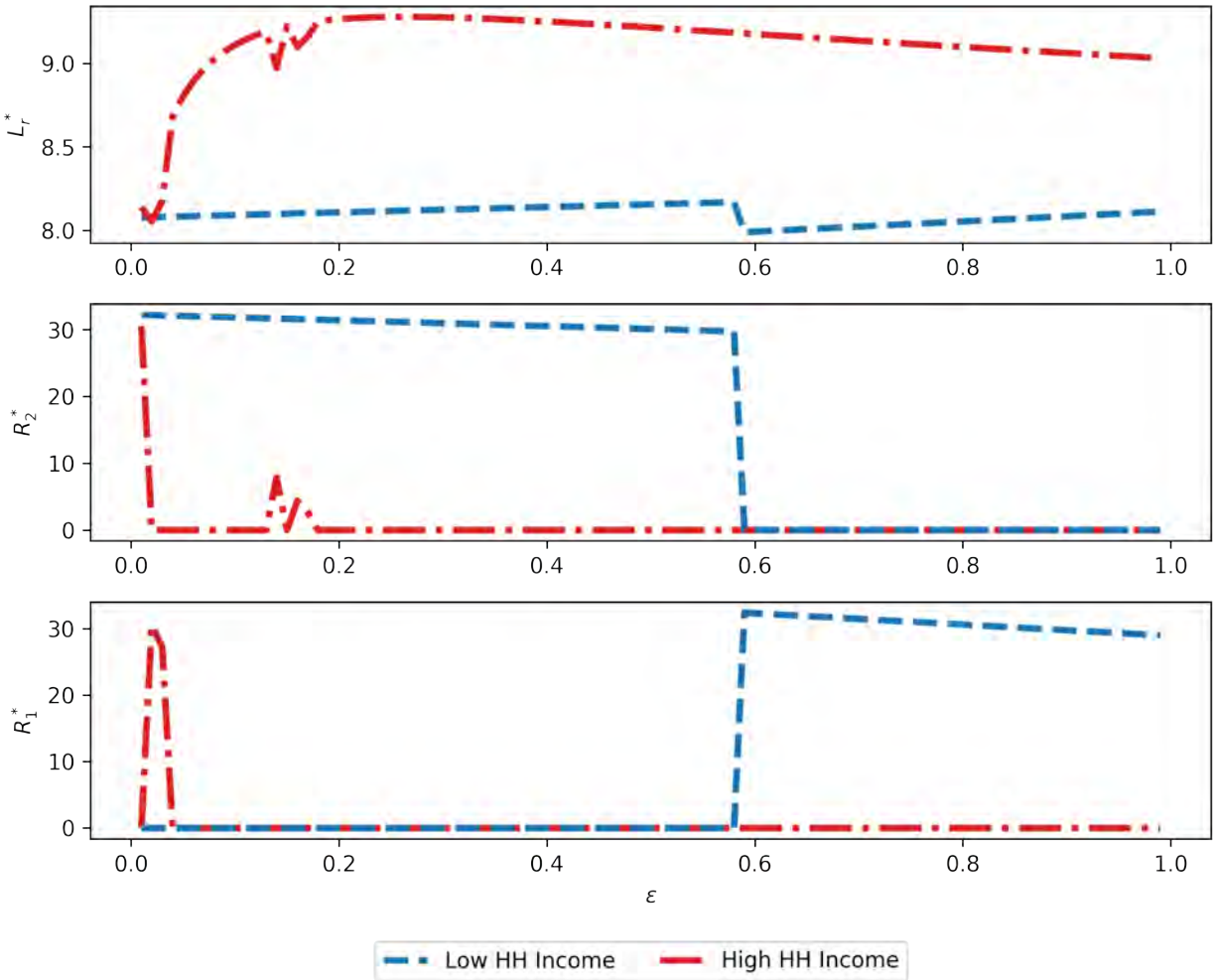

Figure 1: Effect of First Period Shock (ϵ) on Migrant Decision

Note: Figure depicts results of simulations varying ϵ for the migrant.

On the other hand, if we vary α, the results differ. α directly affects production in the second period. This can be due to flood, drought and other infrastructure damage that would make it more difficult to take part in production. Figure **??** shows the optimal choices made by the migrant for changes in α. Compared to the results in Figure 1, the migrant never chooses $R_1 > 0$. Most of the aid is allocated into higher levels of L_r, with the less poor household also choosing to send remittances in the second period.

I can translate these to $L_{1,2}^{p*} - t$ space to see how the driver reacts across the two periods. Figure 2 shows both optimal labor choices for ϵ and α plotted over time. Changes in ϵ are plotted with darker colors denoting higher ϵ and the solid line denoting the poor household. In Panel I, the less poor household's migrant almost always has a decreasing trajectory across time, reducing labor time in response to providing more on-farm labor, and working less than baseline. For the poor household, depending on the level of ϵ, they may either increase or decrease their labor supply across time. Only in the case of a poorer household migrant, is there a positive labor response to a wealth shock.

For Panel II, the fact that $R_1 = 0$ and $R_2 > 0$ for all levels of *alpha* leads to baseline migrant labor time in the first period. In the case of the poorer household's migrant, the migrant chooses to work more and send back R_2 while in the less poor household, there is a higher amount of L_r allocated to the household. This is in line with empirics, where poorer migrants are often constrained in being able to return home too often. The opportunity cost of lost wages is too high, leaving it less costly to send remittances in the form of cash.

In reality, shocks are some combination of ϵ and α. For instance, a drought can be thought of as a high ϵ, low α event: it does not wipe out wealth immediately, but it hinders future

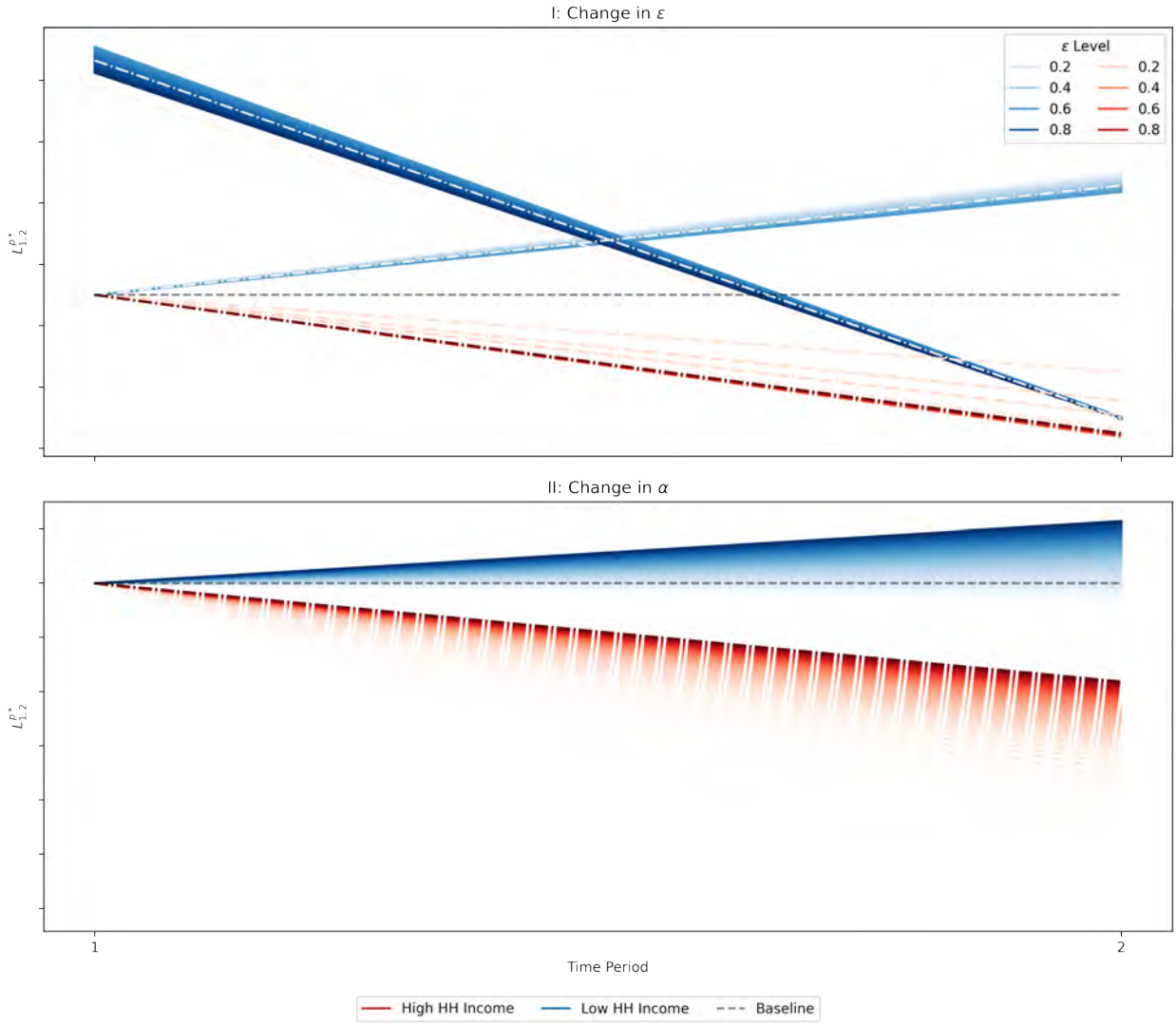

Figure 2: Optimal Labor Allocations across Time for Varying ϵ and α

Note: Figure depicts results of simulations varying ϵ and α, with each line a labor trajectory. Lighter colors denote a lower parameter.

harvest. A flood on the other hand is low ϵ, low α, as it destroys a household's wealth such as houses, lives and capital inputs, and also makes it more difficult to produce yields. The results above illustrate the theoretical ambiguity at play when it comes to observing labor supply decisions in response to rural weather shocks. The exact mix of remittances and labor is not easily disentangled, but the results of the model do show that the severity

of the shock and the wealth of the origin household, have a more pronounced effect on the migrant's labor decisions.

3 Background and Setting

The first subsection starts by describing Uganda's economy, which has traditionally been based on agricultural production, most often at the subsistence level. I then discuss why the country experiences much climate variation and other factors that affect agricultural production. I then turn to the role that internal migration and remittances play in the economy. Finally, I discuss Uber's 2016 entry into Kampala, Uganda's capital city, and the characteristics of a representative Ugandan Uber driver. The second subsection briefly describes how ethnic and clan identities in present-day Uganda derive from the former regional kingdoms. Ugandan surnames have linguistic features that can be used to trace an individual's regional origin.

3.1 Uganda's Economy

Uganda's economy heavily depends on agriculture. In recent years, subsistence agriculture has accounted for 42.6% of income for Ugandans. Among Ugandan households, 32.3% receive their incomes on a seasonal basis from subsistence farming, while 26.6% receive their income daily, mostly through non-agricultural enterprises/employment. The majority of Uganda's poor residents live in rural areas in the county's eastern and northern regions. Along with a widespread dependence on agriculture, Uganda experiences large variability in weather, both

spatially and temporally, due to multiple geographic features such as inland water bodies, complex topography (Ogwang, Guirong, and Haishan 2012), and effects from El Nino (Kovats et al. 2003). Thus, despite its small geographic size, Uganda is a country with significant climatic variation. In terms of agriculture, Uganda often endures several types of adverse weather events that have the potential to hinder crop production and destroy yields, such as drought and flooding (Ogwang, Guirong, and Haishan 2012). But agricultural income is also threatened by soil infestation and plant disease, which are leading causes of agricultural yield reduction, an event that could induce the need for supplementary income (Sundström et al. 2014). These two latter factors can be independent events or secondary effects of climate change (Burzyński et al. 2019). In this paper, I will use a measure of biomass growth across the growing season, which will allow me to capture any event that would significantly hinder crop production in an area.

One of the major pathways out of poverty for Ugandans is internal migration, which is mostly undertaken for economic reasons (Poverty Reduction and Economic Management 2006; Mukwaya et al. 2012). A recent study using the 2016/17 Uganda National Household Survey found that internal migration was mainly undertaken for income-related reasons (39.7%), with over 30.8% of Ugandans migrating to Kampala, the capital city (UBOS 2017). Migrants mainly relocate to the districts of Kalangala, Kampala and the peri-urban district of Wakiso (Mukwaya et al. 2012). Most migration is intra-regional, but Kampala has a large amount of in-migration, with roughly the same percentage of the population that migrate from within the same region (Buganda) as outside of it (27% to 24%, respectively).

Remittances sent from internal migrants are important to the economic life of many Ugan-

dans, and comprise an informal social safety net that alleviates rural poverty. According to the Ugandan Bureau of Statistics, remittances made up an average of 6.5% of income in 2012/13, with several sub-regions having a larger proportion such as the central subregions (8.1% for both) and the Kigezi sub-region (8.8%). In the 2016-2017 period, when rural poverty rose, remittances increased to make up 6.9% percent of rural income; some rural sub-regions relied on remittances for 8-10% of their income (such as the central regions and the West Nile region). As I will go on to show, the predicted ethnic composition of Uber drivers largely coincides with this make-up of remittance senders.

According to the Uganda of Bureau of Statistics, the number of people in the informal labor market as a fraction of all Ugandans employed in non-agricultural activities has been steadily growing since 2010 (from 67% in 2010 to 87% in 2018). The informal sector permits workers to earn money, but through jobs that have little social protection. Despite this drawback, the informal sector generates a large amount of household income and absorbs unemployed workers from the formal sector (Mugoda et al. 2020). However, measuring output from the informal sector has always been a challenge and suffers from either sampling errors, self-reported measurement error (Joshi, Amoranto, and Hasan 2011), or recall bias (Liedholm 1991). This is because more often than not, the measurement of informal enterprises requires household surveys.

One potential solution to this measurement problem has been the wide-scale adoption of mobile phones and the entrance of online platforms to developing countries like Uganda. Over 68% of Ugandans own a mobile phone, and the emergence of new technologies has transformed many parts of life. Large parts of the informal sector have now transitioned to

conducting application-based "gig" economy work (Berger, Chen, and Frey 2018; Shokoohyar, Sobhani, and Sobhani 2020), shedding light on a previously difficult to measure sector of the economy. As of 2021 there has been an increase in the number of digital platform start-ups across Africa, ranging from rideshare services to matching domestic workers with employers.

In advanced economies, rideshare platforms such as Uber have had a positive impact on part-time workers in developed countries by providing an alternative way to create income and a way to flexibly supplement existing income on a part-time basis (Chen et al. 2019; Cook, Diamond, and Oyer 2019). Among families that experience job loss, ride-share platforms provide a way to supplement income during times of unemployment or income uncertainty (Farrell, Greig, and Hamoudi 2019; Koustas 2018; Abraham and Houseman 2019; Jackson 2019). In the developing world, however, there is little evidence for whether drivers exploit, or can exploit this flexibility in the labor market. In a developing country, Uber's flexibility could potentially allow drivers to adjust more rapidly to income shocks and provide a way to get around binding constraints, such as production technology frictions that limit workers' abilities to marginally adjust their hours in traditional work environments.

In the summer of 2016, Uber entered the Ugandan market in Kampala. The minimum cost for a ride is about 3,000 Ugandan Shilling (UGX), which is the equivalent of 0.83 USD. Uber has two main competitors in the Kampala market, Bolt (an automobile taxi service) and SafeBoda (a fleet of scooter taxis). According to an Uber background report, a representative Uber driver in Kampala is university educated, unmarried, and tends to be male below the age of 35. Most of the ride-share drivers have transitioned to Uber from

the existing transportation sector. The report finds that Uber drivers in Uganda support multiple people with the income they earn. Although there is no information in the report on who those people are or where they are located, internal migrants tend to come from rural areas (Nzabona et al. 2019) and their remittances tend to flow to rural areas, and are used for food, education, health and housing expenditures (McKay and Deshingkar 2014). Ugandans, similarly, to other parts of Africa tend to garner strong rural-urban ties after migrating and even after moving permanently (Gugler 2002; Barratt, Mbonye, and Seeley 2012)

3.2 Surnames and Regional Identities in Uganda

Uganda consists of about three main ethnographic regions. The Bantu region occupies the southern half of the country, while the Central Sudanic region forms the northwest corner. Most of the upper half of the country is the Nilotic region, but on or near its border with Kenya there are a few pockets that make up a fourth ethnographic area, the Kuliak region. These present-day regions were former kingdoms that were joined through treaties when Uganda was under British colonial rule, beginning in 1894. In 1962 Uganda gained its independence, and in 1963 it became a republic.

These regional identities are further divided into smaller ethnic groups and then into clans. In some cases, Ugandans identify most strongly with their clan origins, and their surnames often connect back to those clans. Even siblings may have different surnames, with those surnames connecting to a clan, not to a family name.[5] In other cases, there are linguistic

[5]See https://blog.caritas.us/blog-import/blog/ugandan-surnames-understanding-whats-in-a-name/ for an anecdotal account of this phenomenon.

features that can identify a person belonging to a particular ethnic group. Anecdotally, there are stories of how it is possible to pinpoint a person's regional origin (with differing levels of geographic accuracy) purely based on their surname.[6] It is important to note that due to the nature of the weather data being used, I will be concentrating on predicting the formal political boundaries of the country, which closely, but do not completely, encompass the traditional boundaries of the former kingdoms in Uganda. For more information on this, please see Section A.4 in the appendix.

The linguistic signal that I capture is only as powerful as the amount of ethnic and language diversity in Uganda. Although the official languages of Uganda are English and Swahili, any language can be taught in schools or used for legislative, administrative, or legal purposes. Linguistic diversity in Uganda is a large topic of discussion in the areas of language planning (Namyalo and Nakayiza 2015; Altinyelken, Moorcroft, and Van Der Draai 2014).

Ethnicity, by itself, is an important determinant of many social dynamics in Uganda and other African countries. Metrics based on concepts such as ethnic fractionalization have contributed to understanding how differences in ethnic diversity can drive voting behavior (Alesina and Zhuravskaya 2011; Alesina et al. 2003), social norms (Algan, Hémet, and Laitin 2016), and public investment (Golooba-Mutebi and Hickey 2016). In the present case, however, ethnicity will not be the main variable of interest, but rather a channel used to connect the labor supply behavior of Uber drivers to agricultural shocks occurring in various parts of rural Uganda. See Bird and Michuda (2022), a paper that uses a similar method to predict voting behavior based on ethnicity.

[6] I only consider surnames in this case since first names can often be biblical names that would muddle the linguistic signal.

4 Data and Classification

This section gives summary information on the main datasets that I use in the paper. It is important to note at the outset that I use Global Administrative Unit Layers (GAUL) to delineate the geography. The GAUL methodology was developed by the United Nations' Food and Agriculture Organization (FAO) and aims to provide the most reliable and consistent spatial information on administrative units for all countries in the world. GAUL layers were conceived to overcome any fragmentation across time when digitizing administrative unit layers globally. These administrative layers are particularly important because these are the delineations that I will use when predicting rural origins for the Uber drivers. They must be connected somehow to ethnic or regional boundaries that Ugandans find meaningful, whether that be through a shared language or a historical connection to a kingdom. I use the Level 2 GAUL for Uganda, which makes up around ten regions that match up fairly well with Uganda's agro-ecological zones. This Level-2 GAUL allows me to align the weather data with the different agricultural regions, and to make plausible matches of the Uber drivers with their home regions. More background can be found in Appendix Section A.4.

To preview the challenge that my classification approach seeks to overcome, I begin with the observation that the Uber driver data and the weather data cannot be merged. Weather data is at the pixel-time level and Uber driver data is at the driver-time level. If I assume that Ugandan surnames are representative of older ethnic/tribal/linguistic ties and that internal migration is not so common as to completely dilute that historic signal, I can use surnames as a proxy for a driver's rural connection. To bridge these two disparate datasets, I use a nationally representative voter registration dataset that includes both surnames and

registration locations.[7] This provides the training data necessary to train a gradient boosting model that can generate predictions on a regional connection, given a surname. Once I feed Uber driver surnames into the model, I can get a set of predicted regional affiliations, as either a categorical variable or as a set of probabilities.[8] It is important to emphasize that I can't know which driver surnames are most prevalent since I do not have access to this information.

Uber's driver data provides high-frequency logs of hours, earnings, and the number of trips, information that is easily compiled and necessary for business records. As such, the Uber data is very limited in terms of containing detailed demographic information. I get around this problem by matching drivers to regions based on their surnames.4 I use an external, nationally representative voter dataset for all of Uganda, which gives the names of voters and the locations where they registered. This data gives me the ability to link Uber drivers to rural regions without needing to rely on traditional survey methods to gather this information. Once matched, I use a weather shock indicator in the rural sector and see if these adverse events have any effect on driver behavior. The model's predictions might be falsely matching drivers with weather events occurring in areas that they are not connected to. This would tend to attenuate the true effect I am searching for. To account for this potential mismatch, I develop a switching regression estimator, inspired by L.-F. Lee and Porter (1984) that leverages the gradient boosting classification probabilities to account for misclassification. I

[7]The voter registration and weather data were at different levels of spatial aggregation, necessitating the need to aggregate to a level that was common between the two. This was done both for the surname predictions to align with the spatial levels in the weather dataset and because higher levels of specific spatial aggregation lead to better prediction accuracy.

[8]The surnames in the voter registration dataset were publicly available, but due to the data user agreement with Uber, it was not possible to observe the surnames of the drivers. As such, I created a pipeline where the trained model was transferred to an Uber data scientist and they would generate the predictions without me ever having access to the surnames.

now turn to reviewing the three main datasets in turn; namely, the Uber driver data, the weather data, and the voter registration data.

4.1 The Uber Driver Dataset

The Uber dataset provides complete information on the Ugandan drivers working in Kampala from May 2016, when Uber first started operating in Uganda, through December 2019, with drivers joining the platform over time. It is an unbalanced panel of 14,455 drivers across 189 weeks and is comprised of earnings and hours online spent on the platform, as well as some basic demographic characteristics. The week of a driver's first trip is given, but it is not possible to know if a driver has left the platform. Hours online for Uber can be defined as the time from turning on the online application and becoming open to receiving ride requests to the time the application is turned off. In rare cases, some drivers keep their phone on for long periods waiting for a ride request or they might forget to turn off the application when they get home. This might create the illusion that a driver is online for 24 hours at a time. These cases are dropped from the sample. I define a driver as being on the platform if the date under consideration occurs after the week of their first trip and before the last time they turned on the application. If there are gaps in between these beginning and end points, I assume that their hours and earnings were 0 for that particular week. There is a small drop in the number of drivers after April 2019, as I only fill in observations for drivers between their earliest and latest observations of positive hours. Hours online and earnings are trimmed to the 99% percentile on the weekly level to reduce the chances of outliers affecting the results.

Table 1 shows summary statistics for Ugandan Uber drivers. Average hours online are around 29 hours a week, meaning that many drivers utilize Uber as their full-time job. Only about 1% of the sample is made up of female drivers. Shared vehicles denote the proportion of drivers that share their cars with other drivers; this accounts for 6% of the sample. Weekly earnings are around 234,000 UGX (65 USD) which amounts to about 834,000 UGX (232 USD) a month. In comparison to other labor market opportunities, Uber drivers make as much as highly educated members of Kampalan society. According to the Uganda Bureau of Statistics, in 2016, the median monthly earnings for Ugandans with a college degree who were living in Kampala was about 780,000 UGX (217 USD). This comparison seems substantial, but it is important to note that the income derived from Uber is much more precarious than wage income. The standard deviation illustrates that there is significant variation in mean incomes across drivers and time. Also note that tips are not a significant part of income for drivers. In contrast to Schmidt (2018), where tips are used as earnings shocks, Ugandans do not tend to tip their drivers. More information on vehicle sharing and gender differences can be found in Appendix Section A.1.

Figure 3 shows a plot of mean hours per driver within four-hour intervals spent on the platform, as well as earnings per driver and an earnings index provided by Uber. The earnings index is meant to be a proxy for surges and other platform-specific factors that affect the platform. Mean hours show that as expected, most drivers do not drive within the early morning or late night hours. Most driving takes place between 10 AM and 4 PM. There was a large increase in mean earnings, but this fell as more drivers entered the platform. There are events during the time of the sample when late-night hours (10 PM-4 AM) jumps,

Table 1: Summary Statistics of Ugandan Uber Drivers for Monthly Level

	Mean	Std. Dev.
Weekly Earnings	234,506.19	154,953.80
Tips	75.26	210.71
Age Group	28.38	7.53
Lifetime Completed Trips	1,164.90	1,157.42
Hours Online	28.82	16.72
Female	0.01	0.12
Shared Vehicle	0.06	0.17

Note: Sample standard deviations in parentheses. Age, Lifetime completed trips, and Female are driver-specific. All other variables are driver mean across time. Earnings in local real 2017 Ugandan Shillings, with an exchange rate of 3,600 UGX to 1 USD.

which suggests that when drivers need to increase hours, they tend to increase into off-peak hours, which according to the earnings index, is consistently higher than earnings indices for other chunks of time.

One facet of Uber's labor supply is how consistent drivers are in their behavior. The advantage of some flexible work arrangements is that the individual may have the option to stop work when needed given other constraints or demands on their time. Figure 4 shows a histogram of the number of consecutive weeks that a driver stays off the platform. The mode of the distribution is that a driver stays on the platform for a week before returning. A far smaller number of drivers might come back after 3-4 weeks, which may be due to other labor opportunities or needing to leave because of some adverse event. This illustrates that while drivers might use the flexibility of the platform, they very seldom leave the platform, or if they do, it's usually for a week or two. However, spells of absences may increase in response to adverse shocks. Drivers may leave the platform for short periods to travel back

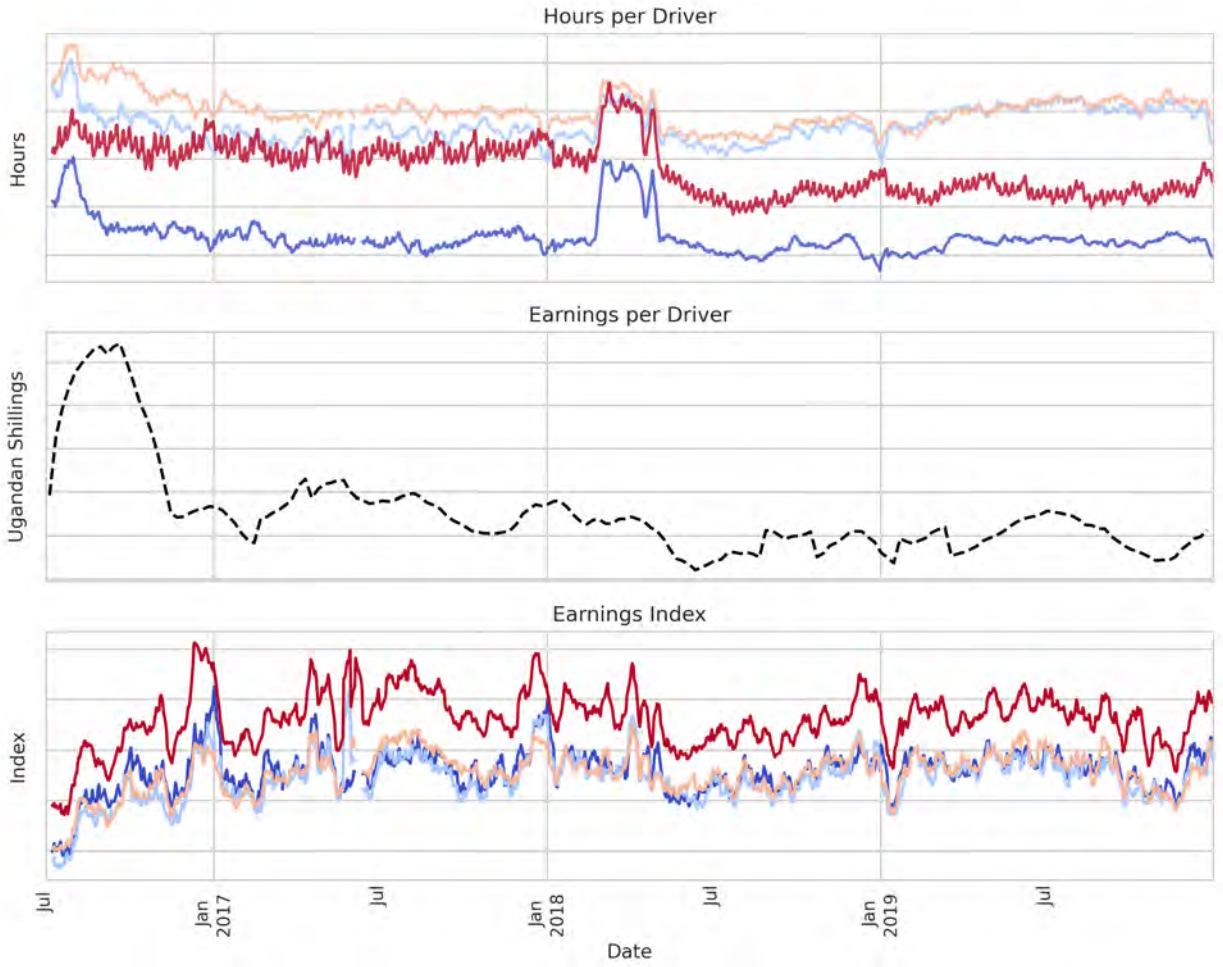

Figure 3: Within Hour Uber Driver Behavior in Terms of Hours, Earnings and an Earnings Index

Note: Plots at weekly level, with a six-week rolling mean transformation to reduce noise. Where available, multiple plots denote six-hour intervals of time. The earnings index is defined as the average earnings per hour in a chunk of time, indexed at 1 at the first date observed, 4 am July 1st, 2016. Earnings in real 2017 Ugandan Shillings, with an exchange rate of 3,600 UGX to 1 USD. An active timeline for a driver is defined as all periods between the first and last time the driver was observed. Y-axis labels omitted due to data user agreement.

to relatives or owned land after negative agricultural income events. The possibility of such events will be discussed later in Section 7.

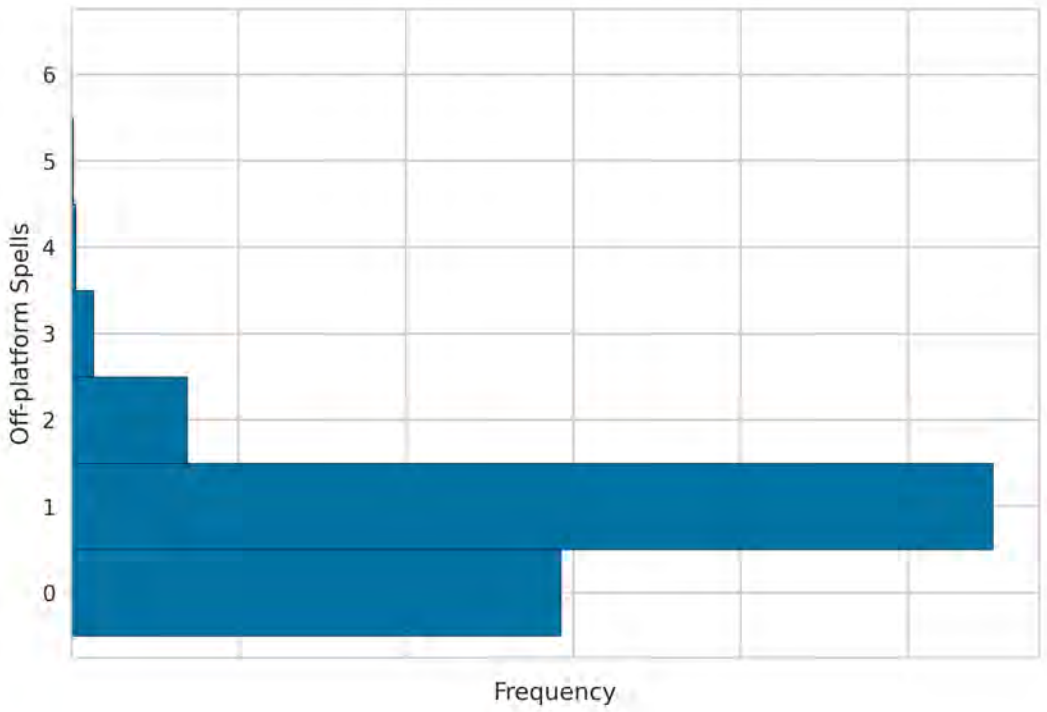

Figure 4: Frequency Graph of the Average Number of Off-Platform Spells for a Driver

Note: Figure depicts histogram of the number of times a driver stays off the platform at a time or a "spell". A spell is defined as the number of consecutive weeks that the application registers 0 hours online for the driver.

4.2 Weather Indicators

I use a rich dataset of weather indicators aggregated by the Anomaly hotSpots of Agricultural Production (ASAP), an online decision support system developed as part of the Monitoring Agricultural ResourceS (MARS) initiative at the European Commission's Joint Research Centre (JRC). ASAP provides a categorical warning system that tracks areas in the world that are at risk of agricultural drought events (Rembold et al. 2019), but they make all constituent parts of their warning system available. The monitoring system uses

various weather indices including temperature, precipitation, and various NDVI (Normalized Difference Vegetation Index) metrics, and utilizes remote sensing to ascertain the phenology of growing seasons and land cover masking of cropland versus range-land. This phenology is then mapped to FAO crop calendars to ascertain the crop being grown at that particular time. Understanding the timing of growing seasons is particularly important, as my approach performs a significant aggregation of spatial pixels to the regions of origin that I predict for the Uber drivers. ASAP data can filter out the weather information that is pertinent to the question at hand. I am able to restrict my analysis to particular growing and harvest seasons to get the best signal of agricultural drought and flood shocks. The data that is provided by ASAP is at the 10-day (or dekad) level and uses the FAO Global Administrative Unit Layers (GAUL) dataset (GeoNetwork 2007) for pixel aggregation. Broadly speaking, there are three types of weather indicators in the ASAP dataset: precipitation-based measures; the normalized difference vegetation index (NDVI), which indicates a drought; and harvest indicators.

For this paper, I use the cumulative NDVI z-score (cNDVIz) over the growing seasons of the crops in each region. This is an index that has gained popularity for its ability to capture the cumulative effect of vegetation growth over the growing season (Ricotta, Avena, and De Palma 1999). The advantage of using this metric over the composite metric developed by ASAP is that NDVI-z score measures the earth's relative "greenness" areas, which signifies crop growth. This level of "greenness" can decline due to drought (which is ASAP's main goal) and other factors, such as flood, disease, or pest infestation. Since I am most interested in crop yields as these pertain to agricultural income, using the cumulative NDVI metric

captures all types of threats to agricultural income that might occur. Please see Appendix Section A.2 for a closer look at the ASAP data.

Figure 5 shows the cNDVIz across the ten GAUL layers for Uganda. The cumulative NDVI z-score (using –1.2 as a cutoff for a severe adverse biomass event) yields 4 events of note: from the end of 2016 to the beginning of 2017, the middle of 2017, the beginning of 2018, and the middle of 2019. Shocks are correlated to one another, ranging from a correlation coefficient of 0.3 to 0.8, with geographic distance being the primary driver of correlatedness. The shaded areas denote events when severe adverse weather events occurred in one or more regions, but not in others. This figure illustrates my identification strategy: that there are times when some Uber drivers are "treated" while others are not. These adverse weather events are largely consistent with events that are ascertained through retrospective shock questionnaires in the Living Standards Measurement Study (LSMS) surveys in 2016, 2018, and 2019. A calendar of retrospective shocks can be found in Appendix Section A.2. Shock events are given by the weather indicator; and the months that shocks occurred, as indicated from the retrospective questionnaire, are not completely aligned with the weather data. This discrepancy may be due to retrospective recall bias, but it may also be because the weather indicator only tends to pick up the damage caused by a shock once it is significantly high, so the weather indicator may report a shock after households on the ground have already detected it.

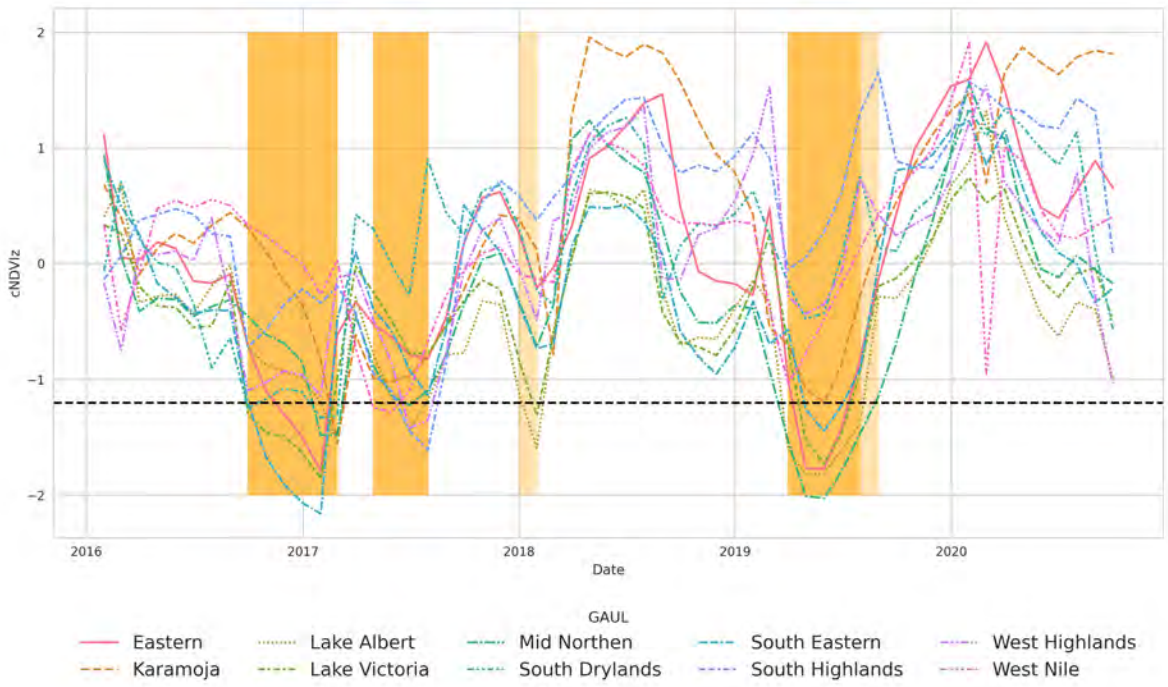

Figure 5: Weather Indicators with GAUL Regions

Note: Weather indicators from https://mars.jrc.ec.europa.eu/asap/, and grouped by GAUL regions. Shaded regions denote times of biomass stress in one area and not in another, illustrating spatial variation. A shock is defined as a Cumulative NDVI Z-score of less than -1.2. Lighter bands illustrate a cutoff of -0.9. Cutoff was chosen based on @rembold2019asap.

4.3 Voter Registration Data

The data used for classifying individual Uber drivers according to their rural region of origin is a nationally representative dataset of voter registration for 2016. It includes information on where Ugandans registered and their surname. I constructed a gradient boosting model to predict their region of registration based on breaking down features of their surname. This method pick ups on linguistic differences so that I am not just predicting a driver's rural ties based purely on the frequency of a name appearing in a particular region, but also on the specific features of the name. It is also more robust to name spelling differences.

Table 2: Summary Table of Voter Registration Data

	District	Surname	GAUL
Count	14,589,191	14,589,191	14,589,191
Categories	112	627,280	10
Most Frequent	Wakiso	Akello	Lake Victoria Crescent
Occurrences of Most Frequent	898,940	103,412	3,442,242

Note: Table shows summary statistics of voting registration dataset. GAUL denotes Global Administrative Unit Layers, as defined by the FAO.

Table 2 shows the summary statistics for the surnames and the labels used in the classification, GAUL layers. There are over 14 million observations in the sample, coming from 112 districts. This accounts for almost all 134 districts in Uganda. There are over 600,000 unique surnames in the dataset, but with repetitions across districts. GAUL layers are the main "target" variable that I classify for the estimation strategy. This was due to a tradeoff between weather shock precision and classification accuracy that is discussed more in Appendix Section A.4.

4.4 Classification of Uber Drivers' Regional Connection

Gradient boosting, and specifically, `xgboost`, the gradient boosting library that I use, has several advantages First, machine learning not only picks up a signal through name frequency, it also picks up a signal based on a surname's linguistic structure. This allows me to pick up connections between surnames that a frequency table would not be able to detect. Second, a gradient boosting approach is flexible and can provide improved prediction accuracy over other algorithms such as naive Bayes or support vector machines. Third, since the machine learning algorithms breaks surnames into constituent parts, the results need not be an exact (or even fuzzy) match of a name. The machine learning model can handle different names,

such as different vowels or name endings. Its disadvantages are its computational expense and the need for significant tuning, which is discussed more deeply in Appendix Section A.4.

In order to make the probabilities computed by the machine learning model align with the actual probability of predicting that a given surname maps into a particular GAUL layer, I also calibrated the probabilities by using isotonic regression, post-fitting (Menon et al. 2012). This method has two advantages: the first is that the predictions are more reflective of the model's actual "confidence" level, and the second is that the calibrator is an ensemble model which increases accuracy. For the target variable, I evaluate the performance of the model by using Group-Stratified K-Fold Cross-Validation. This means that I split the data in such a way such that each "fold" that I trained on preserved the relative proportions of each region (Stratified), and that the testing dataset did not include any surnames from the training dataset, which could inflate performance (Group).

Figure 6 shows a bar graph of the most important features that went into the XGBoost model for one of the cross-validation folds. The other folds were largely similar and are included in the appendix. By far the most important features were for whether the names began or ended with an "a" and whether "a" appeared in the name at all. For the most part, the model was predicting mostly from single letters, however, the 2-gram, "Na," "Ka" and "am" are also important.

Table 8 gives the results of the cross-validation scores. As a comparison, I also trained a random classifier that tries to predict purely by using the most common class in the dataset. We can see that on average, the model has an accuracy of about 76%. Compared to the random classifier, the machine learning model is much more effective and correctly predicting

Table 3: Results Cross-validation Folds of XGBoost Classifier Compared against a Random Classifier for GAUL

	Random Classifier	Calibrated XGBoost Classifier
CV Fold 1	0.160	0.657
CV Fold 2	0.160	0.660
CV Fold 3	0.160	0.667
CV Fold 4	0.160	0.665
CV Fold 5	0.160	0.661
CV Fold 6	0.160	0.675
CV Fold 7	0.160	0.653
CV Fold 8	0.160	0.668
CV Fold 9	0.160	0.657
CV Fold 10	0.160	0.671
CV Fold Mean	0.160	0.663

Note: Table shows results of stratified group cross-validation for a calibrated XGBoost classification model using GAUL layers as a target. Dummy classifier that predicts from empirical distribution of target is shown for comparison.

region using only surnames.

5 Empirical Framework

This section outlines the paper's empirical approach and the switching regression estimation. Since the approach uses weather shocks as a proxy for agricultural income loss. I make the following two claims to carry out the analysis: (1) agricultural income's effect on driver behavior is negligible after including adverse weather events in the regression, and (2) there are no other channels through which the weather is affecting hours/drivers' labor supply decisions. Additionally, having only one city in my sample may make me vulnerable to city-specific shocks that I cannot control. However, the fact that the Uber drivers not treated in a given period act as a control group for those drivers that are treated is meant to account

for any city-specific shocks. This is more explicitly derived in Appendix Section 5.1.[9]

But even in the face of a plausibly exogenous shock, my econometric identification strategy is still at risk because misclassification bias may be present. I do not observe the true, say, driver origin, r^j, but rather a noisy signal $\hat{r}^j$, observed with measurement error. There are several reasons why the predictions might be subject to misclassification. There might be some systematic bias having to do with historical migration patterns that would influence my predictions categorizing some drivers in one region over another. In this case, the potential outcomes equations would include the probabilities of misclassification. For the case of only two classes, this reduces to an often studied problem, which leads to attenuation bias (Black, Berger, and Scott 2000; Aigner 1973). In the case of $R > 2$, the attenuation will depend on which classes are more severely misclassified. Estimates of misclassified groups will tend to be closer to each other, hiding their true effect. For example, if we know that a large amount of historical migration took place between Uganda's northern and eastern regions, making the misclassification rates for those regions higher, then the impacts of weather for the North and East will tend to be more similar to one another, thus biasing their respective estimates.

[9]There is also the question of whether demand effects and reverse causality are influencing my results. For instance, if I was not able to match Uber drivers to ethnic groups and regions, I would have to opt for using an interrupted time series design. In this instance, the demand effects would be a real threat to the identification strategy, since it would be plausible to expect that a weather shock could affect both the drivers and the riders. However, since I can divide the drivers and control for both driver-specific and time-specific fixed effects, a threat to the identification would only come from one case, in which riders would either positively or negatively select drivers whose rural region was experiencing a negative weather event. This possibility seems implausible, given that riders do not get surname or demographic information when accepting or rejecting a ride.

5.1 Potential Outcomes of Misclassified Drivers

Suppose that a driver is connected to some region from a set of regions R. For simplicity, assume that there are three regions, and membership is summarized by a set of mutually exclusive and exhaustive dichotomous variables, $r^j \forall j \in R$. Suppose that weather shocks are also summarized by a region-specific binary variable, d_t^j. We can write out the outcome Y_{it} in terms of its potential outcomes as:

$$Y_{it} = r_i^1 d_t^1 Y_{it}^1 + r_i^2 d_t^2 Y_{it}^2 + r_i^3 d_t^3 Y_{it}^3$$

for driver i from region r at time t. A driver is considered "treated" if $r_i^j d_t^j = 1$. Note that $r^j d_t^j$ is not mutually exclusive. Drivers from multiple regions can experience weather shocks in one time period. The statistic that I am after in this context is the average treatment effect.

Following the potential outcomes framework of Rubin (1974) and its extension to multiple treatments in Lechner (2001), identification of the ATE relies on the following assumption:

$$(Y^1, Y^2, Y^3) \perp\!\!\!\perp r_i^1 d_t^1, r_i^2 d_t^2, r_i^3 d_t^3 | C_{it} = c \tag{11}$$

where C_{it} is a set of controls.

When a driver from region j is shocked with an adverse weather event, the drivers at that time not shocked can serve as a control group. This enables me to calculate an unbiased estimate of the average treatment effect on a given regional shock, using observable quantities:

$ATTR_j = E(Y_{it}|r^j d_t^1 = 1) - E(Y_{it}|r^j d_t^1 = 0)$. So if region 1 experienced a shock at time t, $ATTR = E(Y_{it}^1|r^j d_t^1 = 1) - E(Y_{it}^2 + Y_{it}^3|r^2 d_t^2 = 0, r^3 d_3^2 = 0)$.

5.1.1 Misclassification

To illustrate the effect of misclassification, suppose that the econometrician only has access to a noisy signal of r^j, which we will denote $\hat{r}^j$. For illustrative purposes, also suppose that there is only a noisy signal between regions 1 and 2, but not between 1 and 3, i.e. $Pr(\hat{r}^1|r_i^2) > 0$, and $Pr(\hat{r}^1|r_i^3) = 0$. For brevity, let $\pi_{ij} \equiv Pr(\hat{r}^i|r^j)$. We can rewrite $ATTR_1$ as:

$$\begin{aligned} ATTR_1 =& E(Y_{it}|\hat{r}^1 = 1) - E(Y_{it}|\hat{r}^j \neq 1) = \\ & \pi_{11}(E(Y_{it}^1) - E(Y_{it}^2) - E(Y_{it}^3)) + (1 - \pi_{11})(E(Y_{it}^2) - E(Y_{it}^1) - E(Y_{it}^3)) \end{aligned} \tag{12}$$

Note that $\pi_{11} = 1 - \pi_{12}$. If the practicioner were to estimate this using linear regression in the standard way, it would be a weighted effect of the true effect and a false effect stemming from region 2. To de-bias this estimate, it is necessary to get an estimate of the misclassification. Previous research has utilized modified inverse probability weighting (Shu and Yi 2019), or Bayesian techniques (Gustafson 2003). The current paper takes inspiration from the switching regression literature. But in contrast to classical switching regression techniques, which rely on endogenous identification of the assignment probabilities, this approach uses the exogenously determined assignment probabilities that are generated by a machine learning model.

5.2 Correcting the Misclassification

To make concepts more concrete, I only observe the joint probability $f_{Y_t,\tilde{R}|W_t}(y_t,\tilde{r}|w_t)$ and I do not observe true membership r. I assume that r, $\tilde{r} \perp W_t$, which states that r and $\tilde{r}$ are independent of weather, when conditioned on the names in the training dataset. This assumption is plausible in this setting since there is no causal link between ethnolinguistic features in a name and weather. It could be that population dynamics are affected by weather and people with different places of origin will appear in different regions, but that would likely be from weather-induced migration in the past and not within the time window of the study.

This condition implies that the prediction, $\tilde{r}$, is only a function of the name dataset, so $f_{\tilde{R}|W_t}(\tilde{r}|w_t) = f_{\tilde{R}}(\tilde{r})$. The same is true for $f_{R|\tilde{R},W_t}(r|\tilde{r},w_t)$, so: $f_{R|\tilde{R},W_t}(r|\tilde{r},w_t) = f_{R|\tilde{R}}(r|\tilde{r})$. Conditional on the true region of origin, Y_t is independent of the prediction, so that $f_{Y_t|\tilde{R},R,W_t}(y_t|\tilde{r},r,w_t) = f_{Y_t|R,W_t}(y_t|r,w_t)$. I define $f_{Y_t|R,W_t}(y_t|r,w_t)$ as being normally distributed with mean $\varepsilon_t = y_t - \sum_{j\in R} r^j d_{jt}\beta_j$, and standard deviation σ.[10] d_{jt} is a Tx2 matrix with a column of 1's (for a constant) and a column of weather shock statistics for region j, with β_j being a 2x1 vector of coefficients.

I can then write the likelihood for one observation as in the following equation:

$$f_{Y_t,\tilde{R}|W_t}(y_t,\tilde{r}|w_t) = \sum_{j\in R} f_{Y_t|R,W_t}(y_t|r=j,w_t)\cdot f_{R|\tilde{R}}(r=j|\tilde{r})\cdot f_{\tilde{R}}(\tilde{r})$$

[10] Alternatively, you can also have a separate W_{rt} in the conditional distribution, but this was a clearer way to write the distribution and it only involves needing to estimate one parameter for the standard deviation than $|R|$ parameters.

Thus, I have decomposed the joint distribution into three parts that I can observe:

1. $f_{Y_t|R,W_t}(y|r, w_t)$: The outcome as a function of the true region and covariates
2. $f_{R|\tilde{R}}(r|\tilde{r})$: The conditional distribution of the true region of origin, given the prediction. This can be derived from the so-called confusion matrix that gives the joint distribution, $f_{R,\tilde{R}}(r, \tilde{r}, z)$
3. $f_{\tilde{R}}(\tilde{r})$: The predictions from the classifier.

Since $\hat{r}$ is calculated using the voting registration dataset, I can use the training dataset to evaluate the accuracy of the prediction. I can summarize the misclassification in a confusion matrix, where each cell describes the joint distribution between r and $\tilde{r}$. Denote this as $f_{\tilde{R},R}(\tilde{r}, r)$ for some probability density $f_{\tilde{R},R}$, illustrated in Table 4.

Table 4: The Confusion Matrix for the $r = 3$ Case.

	$\tilde{r} = 0$	$\tilde{r} = 1$	$\tilde{r} = 2$
$r = 0$	q_{00}	q_{01}	q_{02}
$r = 1$	q_{10}	q_{11}	q_{12}
$r = 2$	q_{20}	q_{21}	q_{22}

This is calculated post-fitting to evaluate the accuracy of the classifier. I can use this table to derive a conditional distribution, $f_{R|\tilde{R}}(r|\tilde{r})$. Let $\pi_{i,j} = f_{R|\tilde{R}}(r = i|\tilde{r} = j)$. Using the confusion matrix above, π_{ij} must be equal to:

$$\pi_{ij} = \frac{q_{ij}}{\sum_{k \in R} q_{kj}} \tag{13}$$

Remembering the parameters as defined in Equation 13, the likelihood for one observation for $R = 3$ would be:

$$
\begin{aligned}
&l(\beta, \sigma | y_t, y_t, \tilde{r}) = \\
&f_{Y_t|R,W_t}(y|r=0, w_t) \cdot \\
&[1\{\tilde{r}=0\}\pi_{00}p_0 + 1\{\tilde{r}=1\}\pi_{01}p_1 + 1\{\tilde{r}=2\}\pi_{02}p_2] + \\
&f_{Y_t|R,W_t}(y|r=1, w_t) \cdot \\
&[1\{\tilde{r}=0\}\pi_{10}p_0 + 1\{\tilde{r}=1\}\pi_{11}p_1 + 1\{\tilde{r}=2\}\pi_{12}p_2] + \\
&f_{Y_t|R,W_t}(y|r=2, w_t) \cdot \\
&[1\{\tilde{r}=0\}\pi_{20}p_0 + 1\{\tilde{r}=1\}\pi_{21}p_1 + 1\{\tilde{r}=2\}\pi_{22}p_2]
\end{aligned}
$$

where $1\{\}$ is the indicator function and $p_j = f_{\tilde{R}}(\tilde{r}=j)$. This likelihood function mirrors the imperfect separation switching regression estimator from L.-F. Lee and Porter (1984). But in contrast, it does not simultaneously estimate the probability parameters. Those parameters are plugged in after the machine learning procedure is done. This likelihood function for one observation is then used to define the full likelihood, which is then maximized and estimates for β and σ are found:

$$
L(\beta, \sigma | Y_{it}, W_{rt}, \tilde{r}_i, N) = \prod_i^N l(\beta, \sigma | Y_{it}, W_{rt}, \tilde{r}_i)
$$

Simulation results using this estimator suggest that it can more precisely estimate the true effect of β in the presence of misclassification. This was tested using artificially created data with higher dependent variable variance, smaller differences between regimes, and higher correlation between weather shocks.

6 Results

In this section, I will present the results of the machine learning prediction and the regression analysis. First, I will show where Uber drivers are predicted to be from, as well as visualizations that highlight differences across driver behavior. Second, I will look at how driver hours are affected as a result of weather shocks, utilizing the variation across all drivers and calculating an average treatment effect across the whole sample. Third, I will run the same set of regressions on the inverse hyperbolic sine of earnings, as well as an indicator for positive hours in a given week, to check for extensive margin effects. Fourth, I will present an extensive margin analysis where I will show how the probability of giving rides changes with more intense shocks. Finally, I will present results for driver heterogeneous effects across planting and harvest seasons of three major crops in Uganda: maize, beans and millet.

Most of the analysis will use the following distributed lag model, or some variant thereof:

$$y_{it} = \sum_{t \in \{0,\ldots,8\}} \beta_t Shock_{r,t}$$

$$X_{it}\gamma + \alpha_i + \alpha_t + \epsilon_{irt}$$

where y_{it} is one of the above outcomes, for driver i at week t. $Shock$ will be defined as the Cumulative NDVI Z-score, either as a continuous variable or in categories. α_i are driver fixed effects and α_t are date fixed effects. X is a matrix of covariates that may include a constant term, an indicator for being a female driver, and $SharedVehicle_{it}$, an indicator for whether the driver is sharing their vehicle with another driver. The reason for the lags is to capture changes in behavior through time and to understand the time it takes drivers to adjust their

hours in response to weather shocks. I use eight lags so that I can plausibly capture an exogenous supply-side effect, before price changes and income shock have had a chance to transmit to the demand side. Analogous pre-trend regressions using 8-week lags and leads for every specification are also presented in Appendix Section A.7.3. Tables with coefficients for base specifications for each outcome are also available in Appendix Section A.7.

Note that since the outcome of interest is a z-score, a negative coefficient means that treated drivers increase their hours as a result of a reduction in biomass as indicated by the ASAP weather indicators. This is because higher shock intensity is coming from *decreases* in the z-score. To improve readability, results of continuous measures of Cumulative NDVI Z-score will be multiplied by -1, making this measure the "Negative Cumulative NDVI Z-score" and making the higher numbers correspond to a more severely adverse weather event.

6.1 Predicted Regional Connection and Weather Shocks

Once predictions from the machine learning model have been generated, we can see where the Uber drivers tend to come from. Figure 7 shows a map of Uganda with the number of drivers predicted to be from each GAUL layer, annotated with a count of the number of shock events with a cumulative NDVI z-score less than -1.2. Based on this, most drivers are from the Lake Victoria Crescent region, where Kampala is located, and as expected, many drivers are from the GAUL layers surrounding this region, Southern Drylands, Western Highlands and the Eastern region. But there also seem to be a fair amount of drivers from the West Nile region of Uganda, which is in the North of Uganda. In terms of the shock events, it is plausible to hypothesize that more shock-prone areas would tend to have more driver representation

as households with a higher need for insuring against those shocks. However, from Figure 7, we see that although there may be some association, this isn't clear. By far, the largest proportion of drivers on the platform are also those that experience the most shocks, Lake Victoria Cresent. However, this may also be a coincidence as Kampala is located in that region. This illustrates that there are many other factors that could drive entrance onto the platform that do not have a connection to driving being used as an insurnace mechanism; initial income to rent an automobile, or Uber platform ethnic networks may lead to a higher precedence of some groups being more represented than others. However, this does not bar the fact that I have sufficient power for estimation, or that this phenomenon exists, only that there may be other motivations for joining other than using it as a way to buffer against household income shortfalls.

Now that I have generated predictions for each driver, I can analyze driver behavior, within each class. In Figure 8, I plot driver average hours within the Lake Victoria and Mid-Northen regions to illustrate the differences between the drivers. I also condition on drivers for whom I am exceedingly confident in my prediction. What we see is that the Lake Victoria Crescent region exhibits a large jump in average hours in the beginning of 2018, the largest group in the sample. For the drivers from the Mid-Northen region, we see that drivers exhibit very different behavior for hours online, and notably, there is no such spike at the beginning of 2018, which was a drought that only affected the Lake Victoria and Lake Albert Crescent regions.

6.2 Effect of Weather Shocks on Hours Online and Earnings

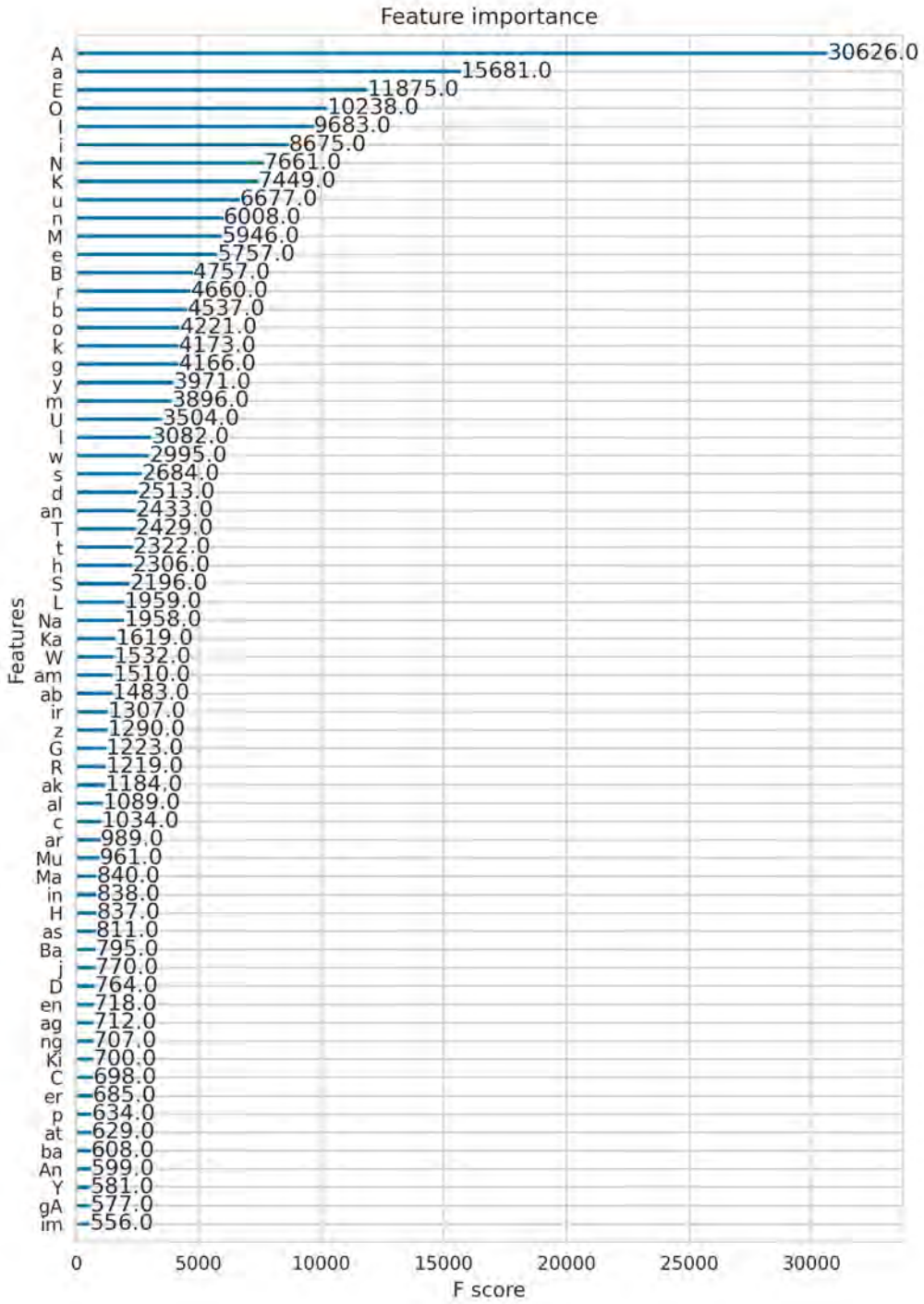

Figure 6: Feature Importance Plot of XGBoost on Surnames Predicting GAUL Layers

Note: Figure shows a bar graph of the 65 most important features, out of 15,255 in the XGBoost Model for one of the folds of the training. Capitalized letters denote the letter appearing either in the beginning or

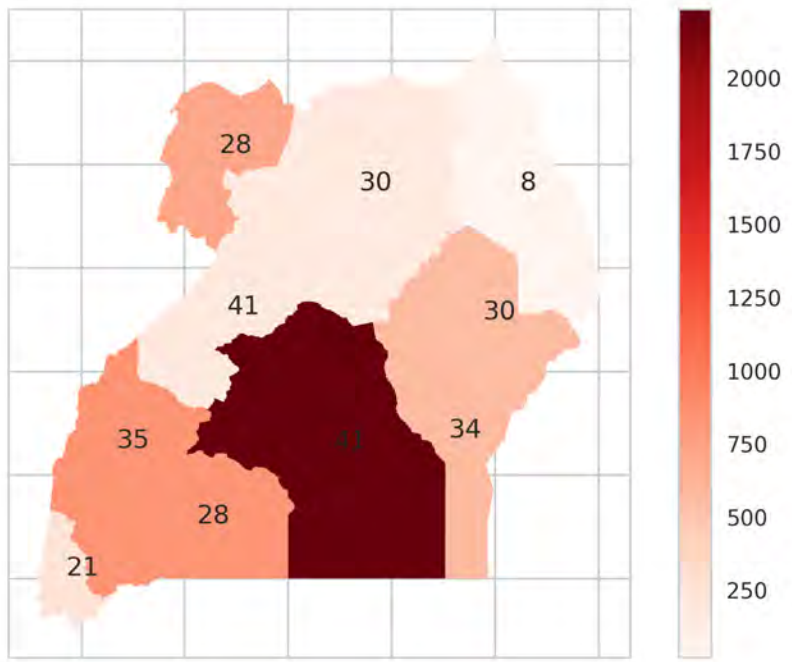

Figure 7: Predicted Uber Driver Origins

Note: Predictions generated from gradient boosting classifier. the lighter the color, the more drivers from that region.

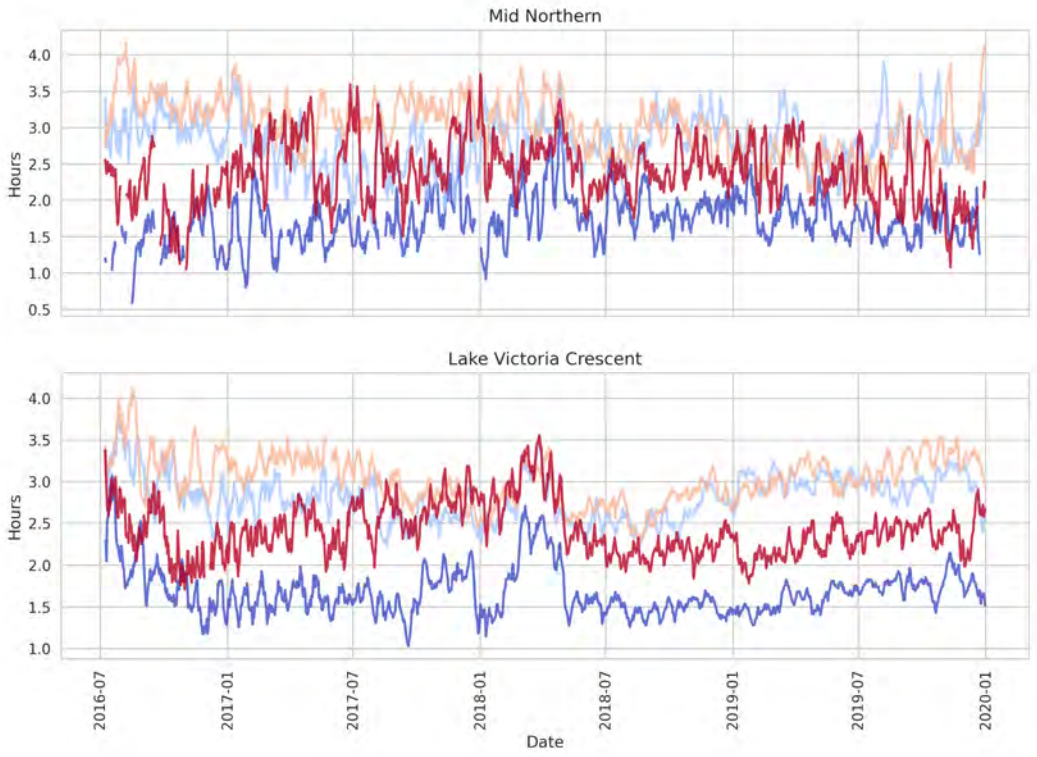

Figure 8: Hours per Driver for Drivers in Mid-Northern and Lake Victoria Crescent Regions

Note: Plots at weekly level, with a six-week rolling mean transformation to reduce noise. Multiple plots denote six-hour intervals of time. Each plot involves subsampling on drivers predicted to be from a particular GAUL region with at least a probability of 0.7. Plot averaging over a mean of 35 and 147 drivers across periods for Mid-Northern and Lake Victoria Crescent, respectively.

Table 5: Effect of Drought Intensity on Hours Online

	OLS			Switching Regression		
	Hours Online	Hours Online	Hours Online	Hours Online	Hours Online	Hours Online
Shock Intensity t	0.393	1.573***	0.924***	0.794*	2.194***	6.704***
	(0.326)	(0.264)	(0.210)	(0.425)	(0.348)	(0.603)
Shock Intensity t-1	-1.250***	-1.110***	-1.598***	-1.763***	-1.835***	-5.181***
	(0.272)	(0.292)	(0.334)	(0.344)	(0.334)	(0.818)
Shock Intensity t-2	0.407	0.713**	0.261	0.381	0.872**	0.563
	(0.294)	(0.278)	(0.314)	(0.394)	(0.347)	(0.490)
Shock Intensity t-3	-0.462**	-0.254	-0.536**	-0.399	-0.187	-0.234
	(0.194)	(0.193)	(0.251)	(0.304)	(0.316)	(0.409)
Shock Intensity t-4	0.688***	0.747***	0.556**	0.949***	0.953***	1.339***
	(0.128)	(0.179)	(0.239)	(0.198)	(0.269)	(0.398)
Shock Intensity t-5	0.195	0.283*	0.029	0.319	0.326	0.047
	(0.149)	(0.148)	(0.159)	(0.175)	(0.192)	(0.346)
Shock Intensity t-6	0.066	0.135	-0.083	-0.032	0.059	-0.129
	(0.262)	(0.280)	(0.297)	(0.320)	(0.326)	(0.390)
Shock Intensity t-7	0.138	0.271	0.207	0.061	0.290	0.331
	(0.284)	(0.292)	(0.330)	(0.379)	(0.351)	(0.425)
Shock Intensity t-8	0.452*	0.608***	-0.023	0.758**	0.839***	0.629***
	(0.255)	(0.135)	(0.165)	(0.263)	(0.117)	(0.187)
Intercept	32.066***			31.971***		
	(0.446)			(0.455)		
Female	-6.243***			-6.243***		
	(1.646)			(1.742)		
Shared Vehicle	-2.664***	-6.227***	-6.083***	-2.654***	-6.231***	-6.202***
	(0.453)	(0.397)	(0.487)	(0.477)	(0.413)	(0.436)
σ				22.231***	17.037***	16.859***
				(0.125)	(0.094)	(0.092)
Goodness of Fit	0.003	0.011	0.005	1104446.486	1049090.356	1046939.354
N	104117	104117	104117	104117	104117	104117
Driver FE	No	Yes	Yes	No	Yes	Yes
Time FE	No	No	Yes	No	No	Yes

Note: Standard errors clustered at the GAUL level. Analysis done with FAO GAUL boundaries disaggregated to 10 regions. "Time FE" refers to date fixed effects, "Driver FE" refers to driver fixed effects. drought intensity measured in terms of negative cumulative NDVI z-score over the growing period of a crop identified by ASAP. Unadjusted R^2 reported for Panel OLS, with Bayesian Information Criterion reported for switching regression models. No intercepts reported for fixed effects models. * p<0.10 ** p<0.05 ** p<0.01

Table 5 shows the regression estimates for the regression on hours online. The first three columns correspond to using a standard fixed effects regression with different amounts of controls and levels of fixed effects. The last three columns give the same specifications but using the switching regression approach. The OLS estimates tend to lessen the true effect, showing an effect size closer to 0. The switching regression accounts for this attenuation and reveals the true dynamics. In the same week as a one standard deviation increase in shock intensity, a driver affected by this shock spends almost around 7 more hours using the online Uber platform. In the following week, the effect becomes negative and the effect size becomes around –5.2: the treated Uber drivers lower their weekly labor supply by about 3.5 hours. These dynamics account for 22 percentage points over average hours in the week of the shock, and then 16 percentage points under average hours in the week following the shock. As time goes on, the effect dissipates and goes to 0 around 4-6 weeks later. The immediacy of the effect is peculiar, but it makes sense in the context of family linkages and mobile phone usage in Uganda. It is plausible that a shock during the growing season can have almost immediate effects on urban labor supply as migrants have strong connections to their rural households, and mobile phones would make relaying such important information less costly.

To trace the change in earnings, I use the inverse-hyperbolic sine of earnings to account for weeks when earnings are 0. The results of the regression with both driver and week fixed effects are show in Figure 9.[11] Similar to the results for hours online, there is a positive effect in the same week as a shock, as earnings increase by 20%. In the week after a shock occurs, there is a decrease in earnings by about 10%. Since I include date fixed effects, this cannot be

[11]Tables similar to Table 5 can be found in the appendix.

due to any macro-economic shocks that affects all drivers. Moreover, these treatment effects only apply to the Uber drivers who have ties to rural areas that were affected by an adverse weather event even occurring at the time that these labor supply changes are observed.

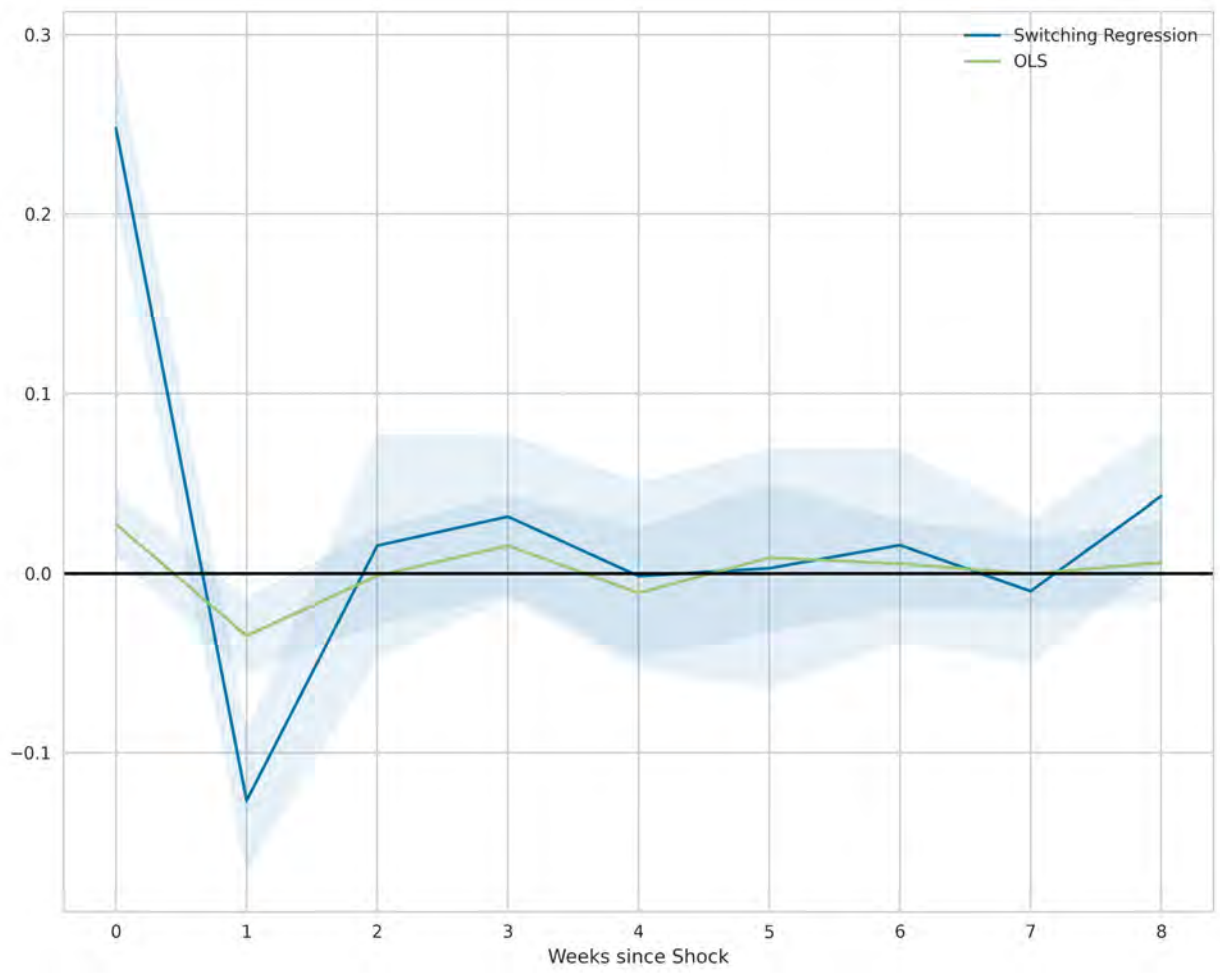

Figure 9: Regression Results for Earnings and Continuous Shock Measure

Note: Figure plots out regression coefficients from a fixed effects panel and switching regression estimation using hours online as dependent variable, of a two-way fixed effects model of lagged measures of shock intensity. Bands show 95% and 90% confidence intervals. X-axis denotes weeks since shock. N=104,117

Another dimension to how drivers react to shocks might be in their heterogeneous responses to different shock severity. However, I do not find significant differences between drivers in that regard. These results can be found in Appendix Section A.7.1.1 and Appendix

Section A.7.1.3 for hours online and earnings respectively.

6.3 Extensive Margin Effects

The results from the previous subsection highlight that driver behavior exhibits considerable changes within just the first few weeks of a shock. One possible explanation for this effect may be that drivers leave the platform shortly after a shock to return to rural areas and help out physically, not just monetarily. This can have important labor market effects if the supply of labor decreases in urban areas as a result of rural shocks.

To explore the extensive margin further, I create an indicator variable for whether there were positive hours in a given week.

To further investigate this effect, I run a linear probability model specification with the switching regression. The results are what we see in Figure 10. The coefficients follow the same pattern as in the results for hours and earnings. There's a large increase in the likelihood of being online (around 40%) and then a decrease a week after the shock of about 50%, which then levels off in the weeks afterward. This decrease in the likelihood of going online may explain the puzzling results for hours and earnings in the previous figures. If drivers are leaving the platform temporarily to return to rural areas after adverse shocks, there would then be an associated decrease in hours and earnings. Additionally, the peculiar dynamics from the above three regressions seem to mirror the descriptive results from Figures 8 and **??**. There are both increases in the same week of a shock, and then stark decreases in the week following. Note that if the extensive margin regressions were only transitory in

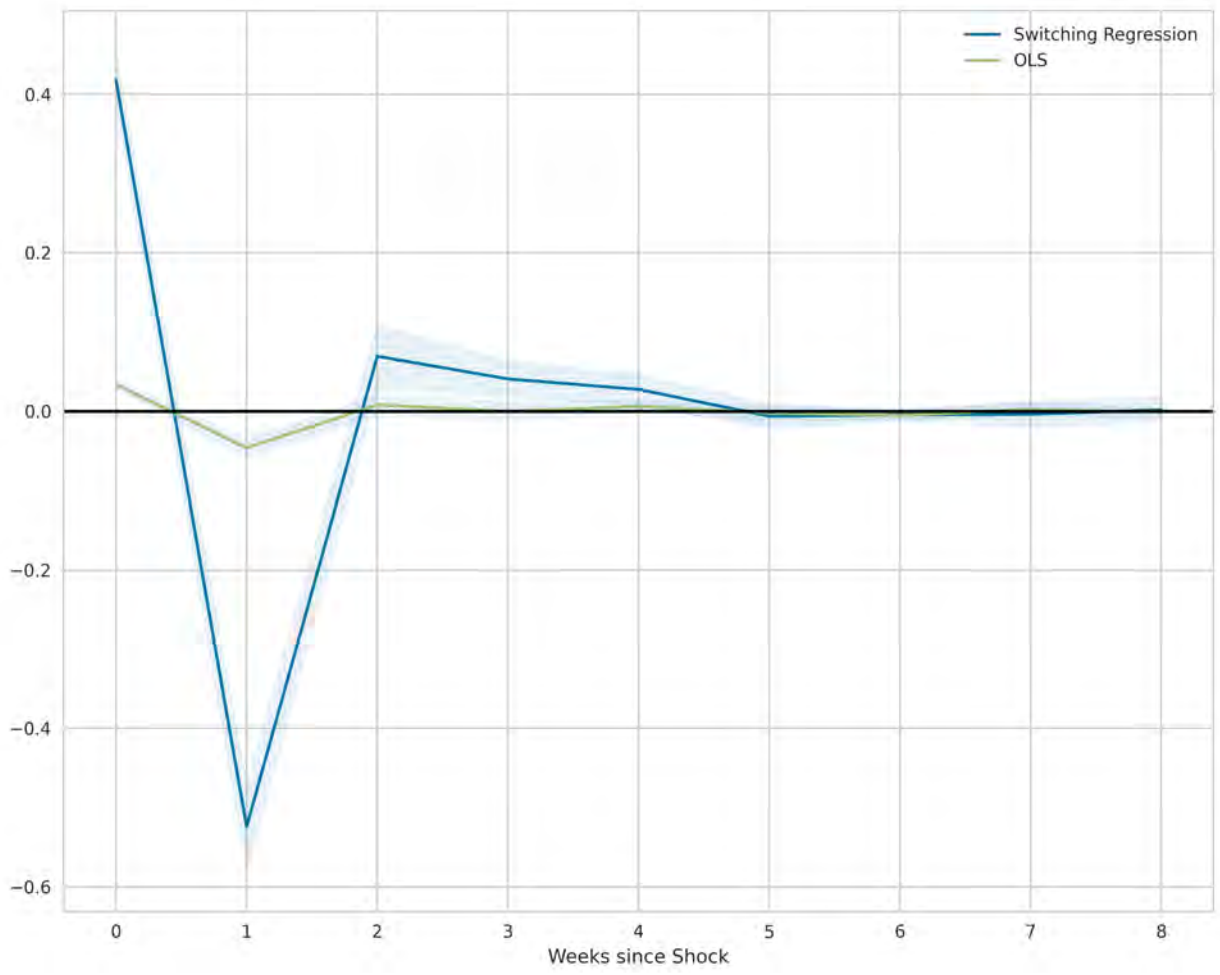

Figure 10: Switching Regression Coefficient Plot of Extensive Margin Effects

Note: This plot shows the coefficients from a switching regression with gender and shared vehicle as control variables. The dependent variable is whether the driver drove positive hours that week. Shaded regions denote 95% and 90% confidence intervals. Standard errors clustered at the GAUL level. N=104,117

nature, then we would expect to see a reversal in the probability of driving in later weeks. However, two weeks after the shock, there is a small increase in the probability of driving, but not enough to account for all drivers returning to work. It would seem that drivers leave the platform and then may not return after a shock, suggesting that they switch to another occupation or leave the city.

6.4 Robustness

Adverse weather shocks affecting Uganda's rural regions seem to induce a labor supply response among Uber drivers in Kampala. Given the online platform's flexibility in terms of scheduling work, the Uber weekly data and the weather data show that the reaction of the drivers affected takes place during the exact same week when the shock occurs. However, it could be drivers are reacting to an urban shock that is not being captured by the week fixed effects, such as general equilibrium adjustments or other changes in demand which I cannot account for. In order to explore this, I will look at whether there is an effect in the time periods before and after the growing season that my shock variables are capturing. Since agriculture is cyclical and seasonal, I will analyze the effect of a shock on driver behavior during the planting season and during the harvest season, which are the two periods of time before and after the growing season.

It may be that drivers react to shock differentially by whether it is planting or harvest season. Since the cumulative NDVI z-score that I use above is only applicable for the growing season of a crop, I use the NDVI z-score instead for this analysis. I run an interaction model with interactions conditioning on whether the shock occurred during the main planting season for the three most important sets of crops in Uganda: maize, beans, and millet.[12] These three crops are planted each year in March. Equation 14 shows this model:

[12] If I use the second season for these crops, I yield similar results.

$$
\begin{aligned}
y_{it} = & \sum_{t \in \{0,\ldots,8\}} \beta_t NDVIz_{r,t} \\
& \sum_{t \in \{0,\ldots,8\}} \phi_t Planting_{r,t} \\
& \sum_{t \in \{0,\ldots,8\}} \psi_t NDVIz_{r,t} \cdot Planting_{r,t} \\
& X_{it}\gamma + \alpha_i + \alpha_t + \epsilon_{irt}
\end{aligned} \tag{14}
$$

Since these three crops are usually planted around the same time, it was not possible to differentiate which exact crop was being planted. Figure 11 shows regression results for this specification. The base effects are shown by the solid line and the effects of hours online at planting (calculated by the delta method) have "+" markers. The confidence intervals overlap with the base effect across all time periods and are not statistically distinguishable from 0. However, for the week after the shock, there is a statistically significant negative effect to hours, which is more pronounced than for the base effect, although not statistically different from it. Shocks do not seem to affect driver bahavior when before the growing season.

I run a similar interaction model, but using indicators for the harvest season for maize, beans, and millet that starts in mid-June and runs through July. The results are shown in Figure 12. In this case, I do see that there is a statistically significant reaction to shocks during harvest, but it statistically distinguishable from the base effect. This is not surprising as, for instance, maize is most vulnerable to a drought during its flowering stage which takes place during the growing period. By the time harvest arrives, the largest damage has already been incurred (Ribaut et al. 2009). As a result, I cannot conclude that drivers react to time

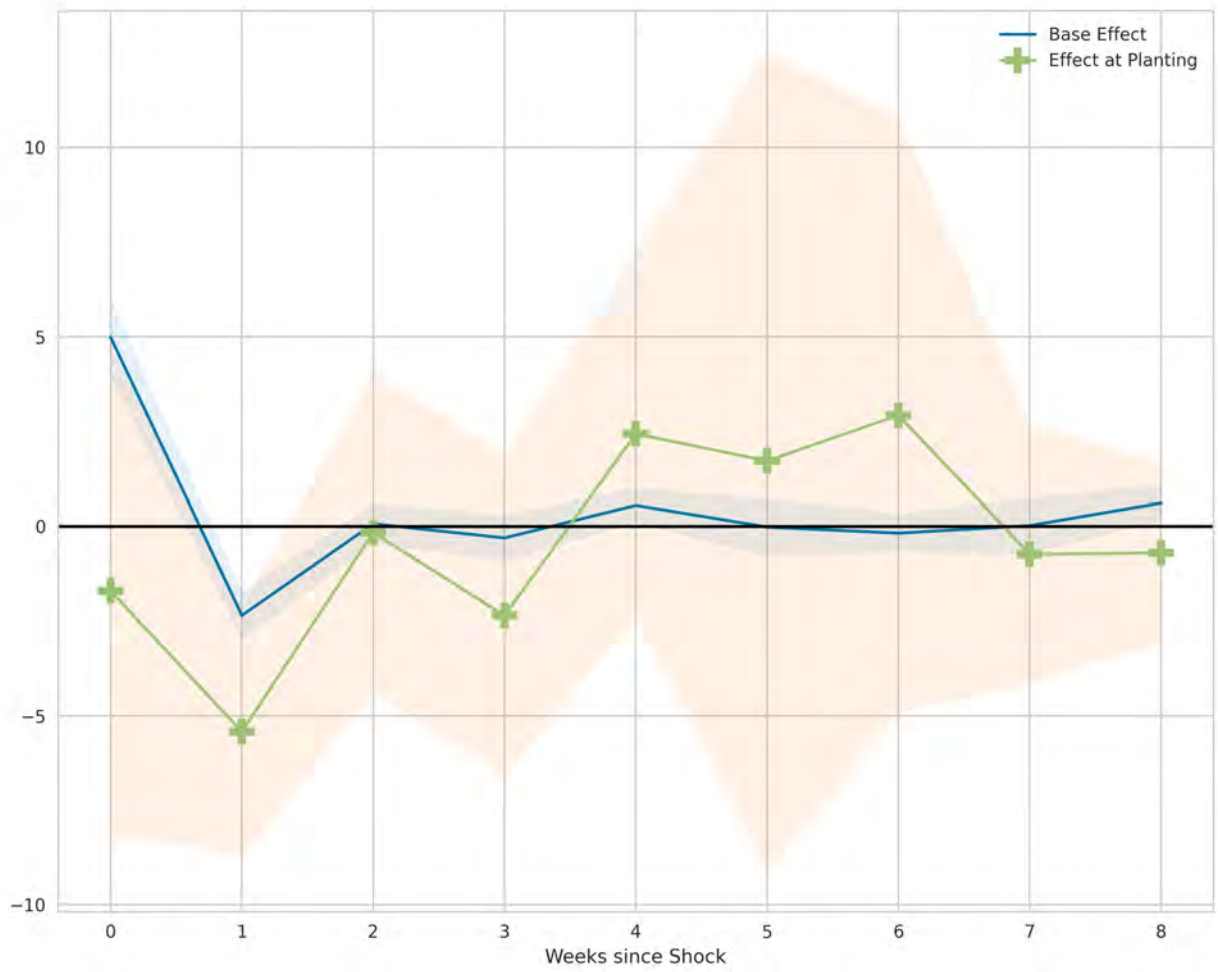

Figure 11: Hours Online during Maize Main Planting Season

Note: Coefficient plot shows the results of an interaction model of each of the distributed lags if also during the main planting season for maize, beans or millet. Shaded regions denote 95% and 90% confidence intervals. Standard errors clustered at the GAUL level. N=104,117

periods that are unrelated to the growing period of the crops of interest.

6.5 Driver Reaction in Times of Favorable Growth

If shocks are affecting drivers in ways other than through the mechanism of helping their household, then it would conceivable that drivers would also react to positive shocks as well.

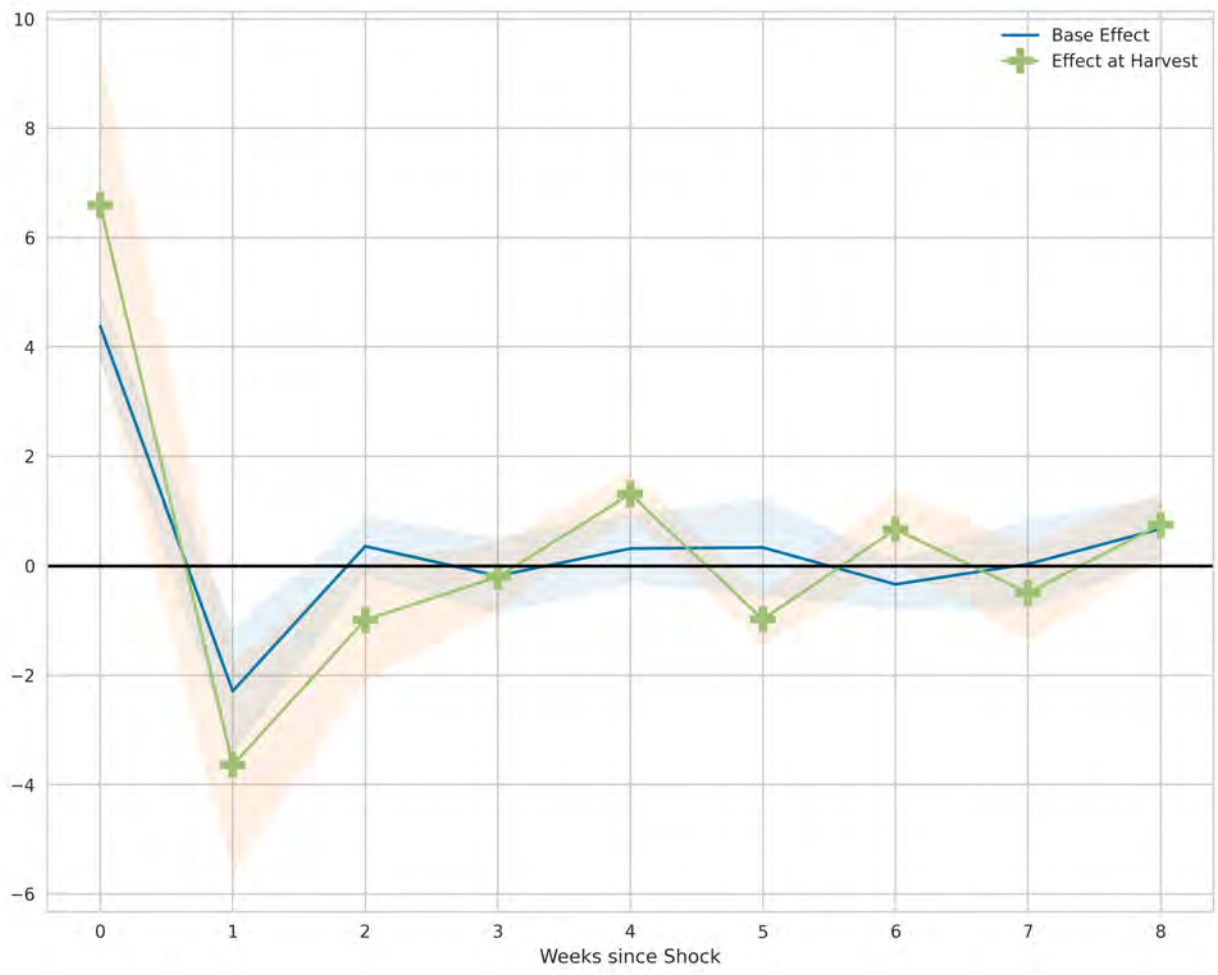

Figure 12: Hours Online during Maize Main Harvest Season

Note: Coefficient plot shows the results of an interaction model of each of the distributed lags if also during the main harvest season for maize, beans or millet. Shaded regions denote 95% and 90% confidence intervals. Standard errors clustered at the GAUL level. Total effects calculated with the delta method. N=104,117

For instance, if the growing period for a season is particularly favorable, and the spatial tranmission of prices turns out to be higher than I claim, then there would be an effect on prices in the city, and therefore urban demand. I will use a cutoff of 1 and -1 for the cumulative NDVI z-scores to capture if drivers react to a favorable or unfavorable shock. Figure 13 shows the results of this regression. The starred line denotes a positive shock, and the dashed line shows an negative shock. The negative shock has the same dynamics as

the previous figures on hours online, and the positive shock shows the opposite effects in the week of the shock and the week after. Apart from a small decrease in hours online one month after the shock, the lagged effects of the positive shocks are not statistically significant.

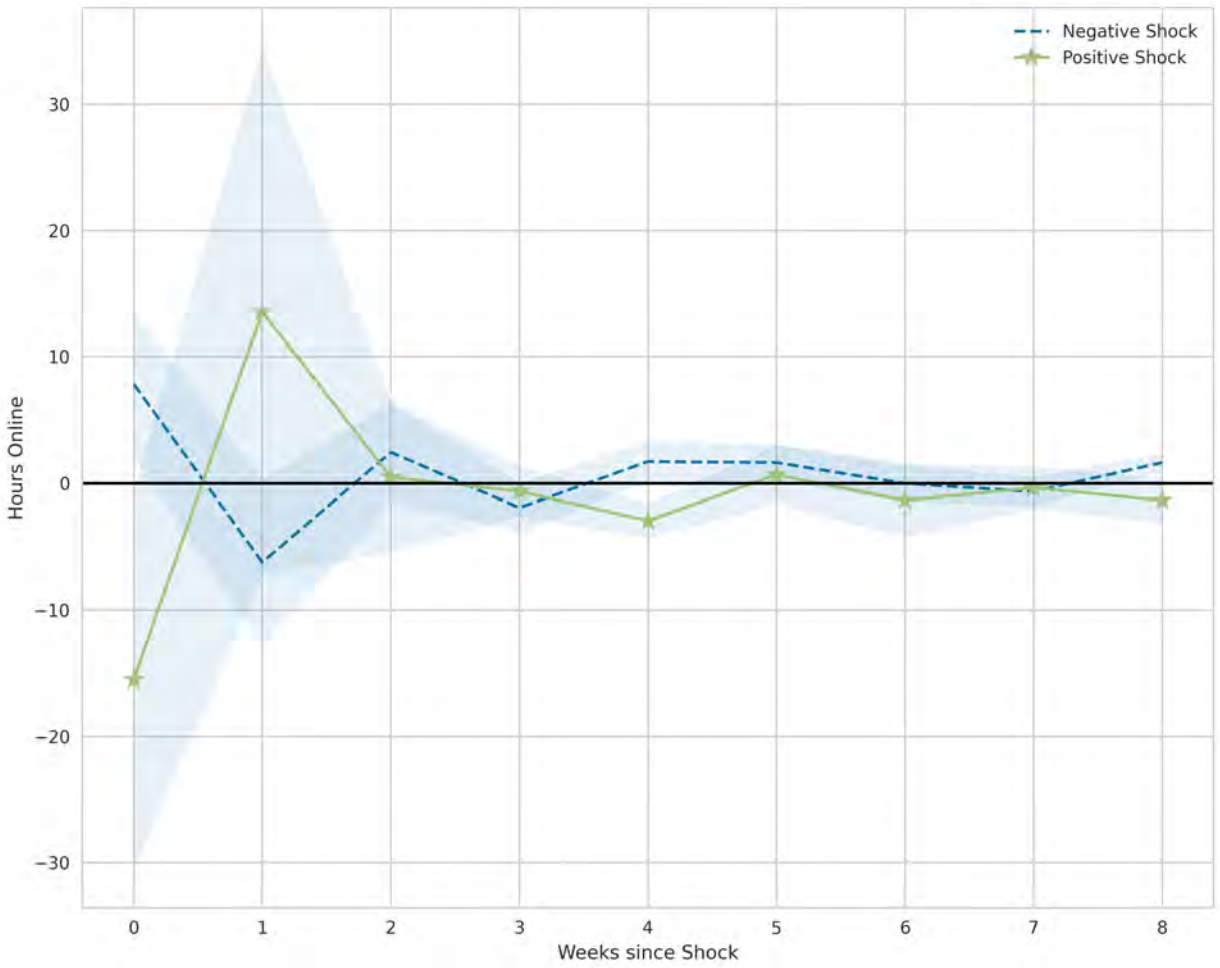

Figure 13: Hours Online during Favorable Growing Conditions

Note: Coefficient plot shows the results of positive and negative shocks to biomass growth, using cutoffs of 1 and -1 for favorable and unfavorable growing conditions, respectively. Standard errors clustered at the GAUL level. N=104,117

7 Discussion

The results suggest that drivers are utilizing the flexibility of the platform, both in how quickly they react to shocks, and that they react strongly on the extensive margin in the week following the shock. Given the identification strategy, the effects that the regression show cannot be due to driver-specific effects or to idiosyncratic common shocks that all drivers face, such as transitory demand shocks. Since the time horizon I am considering is at the week level, it is also unlikely that price changes due to these shocks have been transmitted to the city either. The evidence I provide implies that drivers are reacting in response to income changes that are happening to households affected by these shocks, who are often dependent on remittance income and that may be liquidity constrained. Households that are liquidity constrained may be in need of cash or labor to cope with the shock. This could require hiring labor which cash-constrained farmers might not be able to do, or needing cash to buy inputs. Another explanation might be that to get around cash constraints, the driver is recalled home to act as family labor. This would entail that areas that are more constrained would tend to see higher decreases on the extensive margin. Or that households and migrants react differently depending on the type of shock that is faced.

7.1 Disentangling Floods from Droughts

The extensive margin effect persists in the case of earnings, hours online and even extensive margin hours as well. However, the effects do not seem to be differentiated by planting or harvest seasons. This is plausible as rural households may not necessarily need extra

household labor or physical aid in the case of a shock during planting or harvest. But it may be that the type of shock is crucial to understanding a driver's reaction. In order to explore this hypothesis, I use a weather indicator that can capture both droughts and flood events, the Standard Precipitation Index (SPI). Since the SPI captures an abnormal amount of precipitation against a long-run mean, I can capture both abnormally wet and dry events.[13] As discussed in Section 2, large infrastructure-reducing events can lead to migrant recall. Since floods are more similar to the latter type of shock, I would predict that these events would lead to an increased demand for labor services and an increased likelihood of migrant recall. Thus, flood events could drive the extensive margin effects that I see in the previous sections.

Figure **??** shows a coefficient plot of the results using SPI on the intensive and extensive margins, as well as the inverse hyperbolic sine of weekly earnings. For an abnormally dry event, drivers tend to decrease the probability of being on the platform during the same week as a shock, but their hours on the intensive margin increase. This suggests that drivers may be taking on more work in a different occupation or substituting over to time intervals that are more profitable. In the same week as a dry shock, earnings increase, suggesting that drivers are driving less but at more profitable times. One week after the shock drivers' hours on the intensive margin decreases, but their probability of being on the platform increases. Two weeks after the shock, there is no effect on the intensive margin, but they are less likely to be on the platform. After the week of the shock, however, there is no longer an effect to weekly earnings. In contrast, intensive margin and extensive margin effects are negative in

[13]The reader can compare the cNDVIz and SPI from Figure 18. Note that the SPI may indicate an event *before* the cNDVIz would register a significant decrease in biomass growth.

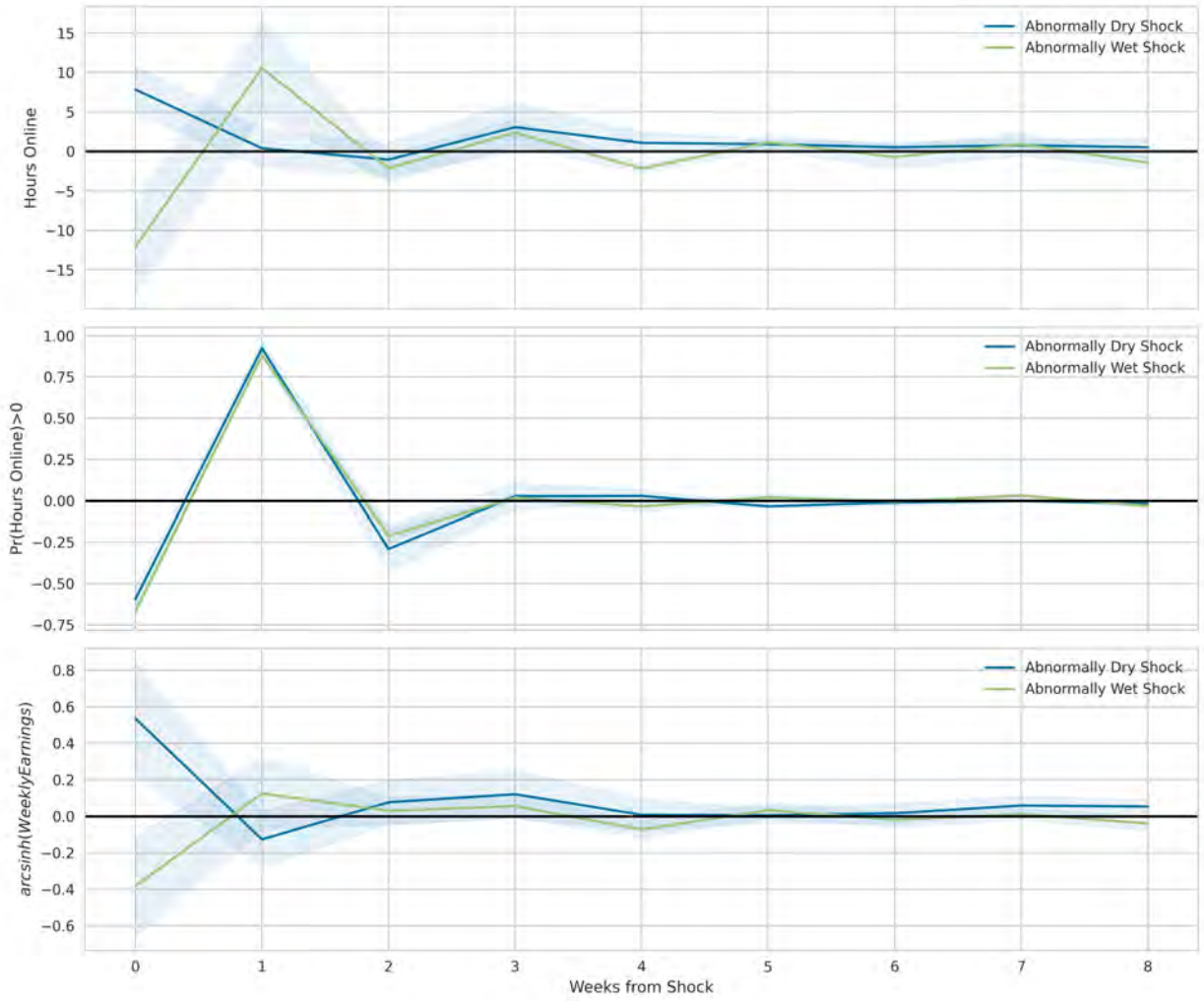

Figure 14: Driver Labor Supply by Type of Shock

Notes: Shock constructed from 3-month Standard Precipitation Index (SPI). Cutoffs for dry and wet shocks were SPI <= -1.2, and SPI > 1.2 respectively. Plot shows the results of an interaction model of each of the distributed lags with each predicted GAUL region. Thick solid line represents effects for the Karamoja region. Shaded areas denotes 95% confidence intervals. Standard errors clustered at the GAUL level. Total effects calculated with the delta method. N=104,117

the week of a shock for drivers that experience an abnormally wet shock. This suggests that drivers leave the platform that week in response to the event. This is further buttressed by the fact that drivers earn less as well. The extensive margin and intensive margin effects are both positive, suggesting that they later return to the platform. These results are in line with the results of Chakraborty and Pandey (2021). In the case of a dry shock or drought, drivers

may take on another occupation or drive more intensively in order to increase their income. But in the case of a wet shock or flood, drivers will instead choose to leave the platform as the value of their labor services outweigh the amount of income they can bring.

7.2 Rural Constraints and Urban Driver Dynamics

To explore how rural constraints may be affecting driver dynamics, I run a regression interacting region of prediction of shock intensity. Figure 15 shows the results of these regressions with total effects for each region plotted using the delta method. All regions tend to exhibit the same behavior during and one week after the shock, with a large increase in the probability of driving in the week of a shock and then a decrease one week after. For most regions, there is then an increase in the probability of driving in the two weeks afterwards. However, driver behavior differs for drivers predicted to be from the Karamoja region. The Karamoja region of Uganda is in the north of Uganda which tends to be one of the least developed parts of Uganda, even amongst the Northern regions, which tend to be the poorest part overall. People in Karamoja are characterized by vulnerability on multiple dimensions, including acute poverty, little protection from drought, limited market opportunities as well as high exposure to the effects of climate change (Mubiru et al. 2010). In 2019 Karamoja ranked lowest in terms of the Human Development Index of all regions in Uganda. In Figure 15, we can see that drivers react differently to the shock, than from other areas. There is a smaller positive extensive margin effect in the week of the shock, and the negative extensive margin effect persists for several weeks. Only four weeks after the shock is there is a positive extensive margin effect. This extended period of being off the platform might be suggestive

of traveling back to Karamoja to help with the aftermath of shocks.

From the evidence above we know that drivers react to these shocks in rural areas and that this is a plausibly causal effect. However, the heterogeneity analysis does show that differences across regions drive differences in behavior and that the flexibility of the platform can allow for different ways of coping with shocks across time. We know for instance, that acute conditions of poverty make it difficult to buffer against shocks; or with a constant need of government aid, remittances or rural-urban linkages become more important in the context of coping with shocks. Based on data from the LSMS-ISA from 2017-2019, I used their shock coping module to construct what households did as the primary way of coping with flood or drought events. Figure 16 shows a bar graph with the results. Karamoja stands out in this bar graph as a region that households reported most in receiving aid from the government, and is also heavily reliant on unconditional help from friends and relatives which can include remittances or some other type of aid. This is most stark when we juxtapose that with the fact that Karamoja tends to be the least to report on relying on savings as the primary coping strategy. Large amounts of unconditional help and little ability to cope with savings helps to explain why Karamoja might have different driver dynamics. Drivers tend to be off the platform for longer spells; and weeks after the shock, there seem to be positive extensive margin effects, which might imply that drivers from Karamoja need to keep providing their families with funds even weeks after the shock.

The true mechanism behind why drivers are reacting this way to weather shocks is unclear. The LSMS-ISA data provides suggestive evidence that savings and liquidity constraints might be partially driving it. What is known, however, is that drivers utilize the flexibility, both

in being able to quickly ramp hours when needed, and to leave the platform temporarily if needed as well.

8 Conclusion

This paper uses a novel and rich dataset of Uber drivers working in Kampala, Uganda's capital city, to show how flexible, informal labor market opportunities are leveraged by households in LMICs to buffer against idiosyncratic shocks in the absence of formal insurance. It is difficult to pin down the exact mechanism at play, but what I find is that drivers do utilize this flexibility and leverage it for a variety of possible reasons to get around credit or liquidity constraints. This paper shows that rural-urban linkages are strong and are crucial in LMICs in helping to buffer against shocks. Moreover, this may also point to a reason why the growth of the informal labor market may not just be driven by unemployment and a lack of jobs. Informal and precarious work gives people an opportunity to shift and adapt to changing circumstances. This is crucial in a context like Uganda, where much of the agriculture is rain-fed and is vulnerable to idiosyncratic shocks. The value of an urban migrant's employment is not just derived from their ability to generate income, but also from how quickly they can change their hours to fit new circumstances.

My approach employs machine learning techniques to overcome previous barriers to connecting datasets used in this analysis. Whereas in the past, household survey data was necessary to understand internal migration, I connect drivers to rural regions using a voter registration dataset and a gradient boosting classification model exploiting the linguistic features of the

drivers' surnames. This methodology allows me to plausibly connect Uber drivers to rural home regions outside of Kampala to quantify how strongly rural weather shocks may affect the drivers' labor supply decisions. But as with such any predictions, there is the potential risk of misclassification which may prevent establishing a causal effect in a regression context. I develop a novel switching regression estimator using the predictions from the machine learning model to be robust to potential misclassification error in the machine learning estimates, yielding largely unbiased estimates of the effect that rural weather shocks in a developing country context have on drivers' labor supply decisions.

This same approach can be used in economic contexts outside of the immediate application in this paper. Machine learning techniques offer policymakers the ability to use existing data (such as names in a census or voter rolls) to elicit information that would allow them to make policy-relevant statements about the economy, without the need to conduct expensive and time-intensive household surveys to gather raw data. This can come with the added benefit of circumventing non-response bias and other sources of selection in sample collection. But it should be noted that while machine learning techniques allow researchers to unite and reconcile once-disparate data sources, this methodology entails a tradeoff: the ability to carry out analysis comes at the cost of incurring prediction errors.

Future work will focus on improving the method by which shock indicators are constructed. There is also much to be done to improve the machine learning model. Most of this paper focused on the misclassification errors that occur across regimes, but more work is needed to understand how misclassification can have a dynamic effect as well. This topic is interesting given how the effects of the lags changed when comparing OLS to maximum likelihood

estimators. Although this paper focused on a machine learning substitute that eliminated the need to conduct household surveys, future work will include investigating the behavior behind the labor market decisions reported in this paper by surveying people in Kampala directly. This will help to better elucidate what incentives and motivations the labor supply effects found in this study.

Given that most migrants to Kampala come from different regions of Uganda, in this case, the population under examination is particularly relevant for the study of economic development and poverty alleviation. In some parts of Uganda, remittances make up/account for more than 10% of a household's income, making a policy targeted at a city potentially important to rural areas as well. If these results are even partly driven by remittance behavior, then the flexible work options that a firm like Uber provides can be a good tool for vulnerable rural households to insure against adverse shocks that threaten their main source of income.

Using a theoretical framework laid out in Chakraborty and Pandey (2021), I find evidence that drivers are more likely to leave the platform in the case of a destructive event that leads to increases in demand for labor services, such as floods. In the case of drought, drivers decrease their probability of being on the platform, but increase hours, suggesting that they are re-optimizing to make more income. For flood events, however, drivers seem to leave the platform, as the value of labor becomes higher than the income they can make as migrants.

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A Appendix

A.1 Demographic Information

Figure 17 shows a density of monthly hours online, broken up by shared vehicle status and gender. Although these two groups are comparatively small to the rest of the sample of drivers, they still make up 808 and 61 shared vehicle and female drivers, respectively. Moreover, these two factors are significant control variables that are used in the regression specifications. Female drivers tend to drive fewer hours on average, which is to be expected, given safety concerns for Uber drivers in general (Almoqbel and Wohn 2019). Interestingly, while there is a large amount of 0-hour observations, there are hardly any for those who share a vehicle. Drivers of shared vehicles tend to have higher hours on average and this is to be expected given the extra variable cost of renting the vehicle. The vehicle-sharing dimension in the Ugandan market is important as it is found more often in a developing context than in the developed world. There is not a large amount of literature on the topic, but for the current context, this extra constraint on the driver might affect a driver's ability to respond to a drought shock, e.g. the driver might not be able to send any money back as most income is going to paying for the vehicle rental and subsistence.

A.2 Weather Indicators

This appendix deals with the weather shock data that I download and process. ASAP provides a categorical warning system that tracks areas in the world that are at risk of agricultural drought events (Rembold et al. 2019), but they make all constituent parts

of their warning system available. The monitoring system uses various weather indices including temperature, precipitation, and various NDVI (Normalized Difference Vegetation Index) metrics, and utilizes remote sensing to ascertain the phenology of growing seasons and land cover masking of cropland versus rangeland. This phenology is then mapped to FAO crop calendars to ascertain the crop being grown at that particular time. Understanding the timing of growing seasons is particularly important in my context, where I undertake significant aggregation of spatial pixels to the regions I predict for drivers. ASAP data gives us the ability to filter out only the weather information that is pertinent to the question at hand. I will be able to restrict my analysis to particular growing and harvest seasons to get the best signal of agricultural drought and flood shocks. The data that is provided by ASAP is at the 10-day or dekad level and uses the FAO Global Administrative Unit Layers (GAUL) dataset (GeoNetwork 2007) for pixel aggregation. Broadly speaking, there are three types of weather indicators in the ASAP dataset:

A.2.1 Precipitation-Based Measures

Cumulative rainfall and SPI are both derived by ASAP from either CHIRPS (Climate Hazards group Infrared Precipitation with Stations) or ECMWF (European Centre for Medium-Range Weather Forecasts). The 3-month SPI is a standard drought index that was developed in McKee et al. (1993) and is a z-score of a structurally fitted time series of precipitation (usually with a gamma distribution). This fitting is done to correct for the fact that rainfall does not exhibit normality at shorter time scales. The SPI, along with the Palmer Drought Severity Index are two standard ways of measuring drought before the advent of remote

sensing techniques like NDVI. NDVI (Normalized Difference Vegetation Index) and SPI are better for different contexts. For instance, according to (Nakalembe 2018), NDVI-based measures perform better than precipitation-based measures in the Karamoja region of Uganda for characterizing agricultural drought.

The WSI is a measure of the availability of water to crops during the growing season. It uses both precipitation and evapotranspiration to estimate the water available to the plant (Boogaard et al. 2018). It ranges from a value of 0 (no rain, dry soil, and complete inability to sustain crops), to 100 (no deficit, and the crop always meets its water requirements).

A.2.2 Cumulative NDVI Z-score

The NDVI is an index using satellite data that measures the level of vegetation in a region, based on land reflectance from satellite data. It measures the level of "greenness" of the earth and the more it turns yellow, the more there is a chance that drought has occurred, as the plant matter on the earth is decreasing. ASAP's NDVI data is derived from the Moderate Resolution Imaging Spectroradiometer (MODIS). For all NDVI-based measures, first, the cumulative NDVI is calculated which measures the increase of vegetation from the start of the season of interest. Then a z-score is calculated using historical cumulative NDVI to get an idea of how typical that pixel's NDVI is that year.

A.2.3 Harvest Indicators

To be more precise about when I posit that a driver will be able to react to a drought event, I also use FAO crop calendars to understand when harvest and growing seasons are. This

way I can better test theories of when a driver would be more likely to react as well as when rural households would be more likely to require assistance.

The three main crops grown in Uganda are maize, millet, and beans. There are two growing seasons, as depicted in Table 6.

Table 6: Harvest Seasons for GAUL Regions.

Season	Crops Grown	End of Planting	End of Harvest
Main	Beans-Millet-Maize	March	Mid June-July
Second	Maize/Millet	Oct.	Feb. (of the following year)

Although there is variation in when growing seasons start, the harvest seasons are usually at the same time. This might be due to market conditions dictating when to harvest, but the end of the growing season and the end of harvest are lined up.

Various theories of information transmission could be applicable in my context. Since the data is noisy by design, we cannot necessarily say what a significant effect is telling us. But if my main hypothesis is to be believed, using the crop calendar in conjunction with the ASAP data can give us a more precise look into why that effect is significant. If a change in hours occurs directly after a bad harvest season, this might stronger evidence for the linkage being driven by agricultural drought.

A.2.4 Comparison of Different Indicators

A.2.4.1 GAUL-SAP Composite Regions and Spatial Aggregation

GAUL layers are a unified set of geographic layers developed by the FAO to overcome any fragmentation across time when digitizing administrative unit layers globally. These

administrative layers are particularly important because these are the delineations that I will use when predicting rural origins for the Uber drivers. They must be connected somehow to ethnic or regional boundaries that Ugandans find meaningful, whether that be through a shared language or a historical connection to a kingdom.

Figure 18 shows a figure of each of the weather indicators for the GAUL-SAP Composite regions. Where applicable, a dashed black line represents a level below which signifies the start of an event where there is abnormally low biomass growth. In Panel A, we see average air temperature, which is negatively correlated with the other three indicators. This is to be expected as higher temperatures are correlated with less precipitation and WSI. Temperature, by itself, however, is not a good indicator as different crops need a certain temperature to grow well. Across time, it seems that there are definite periods that would be candidates for agricultural drought shocks (areas, where z-scores might dip below -1 and temperatures, are high). But what's also important is to find an indicator that can be discerned across regions as well. The benefit of the current approach over, say, identifying on the time series alone, is that when there is enough variation *across* regions, we can identify the effects of weather on hours based on those that are and are not affected by the weather shock. Based on this, however, Panel B for instance would not tend to be a good indicator to take advantage of this strategy as the SPI variables seem to be correlated across regions very strongly. Panels C and D exhibit such behavior: although Cumulative NDVI Z-score and WSI tend to be correlated across regions, there are periods when some regions are doing much better than others.

This can be illustrated by the shaded bands in Panels B-D. These represent periods where

there is a drought in at least one region, with at least one area not being under drought. Compared to Panel B, Panel C has more such periods and those periods are long. Panel D has fewer such events, and while they start at the same times as Panel C, they have different durations.

A.3 Retrospective Shock Questionnaire

In comparison to the weather indicators in the paper, I also used Ugandan LSMS-ISA data to see how much in line my weather indicators would be against real occurrences of shocks. My results show that the Cumulative NDVI Z-score can capture the large shocks that occurred in 2017 and 2019, along with other events that occurred at the end of 2016. Peculiarly, the shock in 2018 was not captured well by NDVI. This shock is probably associated with the East African flood of 2018 which occurred in March-June.

A.4 Details on Classification Methodology and Performance

Matching or predicting a person's ethnic or historical origins is not a new topic of study, albeit it is a controversial one. Surnames have been used since the 19th century to understand relationships between populations subgroups. But there are several issues involved in classifying a person into an "ethnicity" or group, some of which pose a methodological problem and some of which are more philosophical. For instance, the idea of race/ethnicity is often reduced to one variable, but it is part of a multi-faceted set of self-identification. Moreover, ethnic categories are pre-determined based on institutional, historical, or govern-

mental criteria and not based on an open question that is then grouped based on response commonality. "Ethnicity" or any group membership is also something that can change over time based on an individual's self-assessment and it is problematic for it to be determined by bureaucracy or by a computer (Mateos 2007a).[14] However, there is evidence to suggest that there is a correlational chain that connects surnames to rural places of origin. In many places (including Uganda) surnames are passed down patrilineally and are correlated with genetic variation in Y Chromosomes. Moreover, there is also a correlation between variation in genetics and linguistic variation across the globe. In many cases, this can also lead to a significant correlation with geography, although that connection is a weaker one. Several factors can confound the attribution of a surname to a particular geographic location, such as marriage, or migration, but the hope is that there is enough correlation to proxy the rural-urban link.

Various techniques have been used for connecting surnames to ethnic/caste/geography that have been used in social science literature, such as using administrative data and leveraging its representativeness (Algan, Hémet, and Laitin 2016), using context-specific naming conventions (Cruz, Labonne, and Querubin 2017), fuzzy matching search strategies (Mateos 2007b), name classification software using public names data (Smith et al. 2017), probability estimation using frequencies (Bhusal et al. 2020), as well as hybrid approaches using fuzzy matching and a Naive Bayes machine learning method (Monasterio 2017).[15] However,

[14] As my analysis is not centered around finding the *impact* of ethnicity on the variable of interest, but only using it as a proxy for spatial connection, these issues are not as troubling. It is, however, important to note that the impacts that are found throughout this paper are not meant to equate labor market activity to ethnic origins or make any genetic or cultural implications.

[15] Onomap is a software that maps to ethnicity-based on surname data, but is not available for any African country.

these approaches, are often cumbersome to implement well if they were not designed for a particular language, or make assumptions that do not apply to text data (Naive Bayes techniques).[16]

In terms of using a similar exercise but with machine learning, there is robust literature in computer science and computation linguistics around text analysis and language acquisition (Cavnar, Trenkle, et al. 1994), as well as one for using similar and more cutting-edge methods on names (Llitjos and Black 2001; Qu and Grefenstette 2004; Pervouchine et al. 2010). Some early methods included using so-called Naive Bayes techniques or SVM (Support Vector Machines). In general, though, the general conclusion is that these methods work well with phrases and perform worse in the case of surnames.

Moreover, if classification is successful, the estimation strategy would not be successful, if Kampala did not also have a fair amount of ethnic heterogeneity. According to CIA World Factbook, Kampala is a city that includes people of many kingdom memberships, including Buganda (the kingdom in which Kampala resides), Banyankole, Basoga, and others, with significant variation, making the sorting of drivers a potentially powerful identification mechanism.[17]

[16] Fuzzy matching algorithms are often built around a particular language so that it is easier to tell when two words are similar. Since different languages have different common typos or letter substitutions, these fuzzy matching algorithms are very accurate and good at what they do. However, the current paper classifies names from languages for which such fuzzy matching algorithms do not exist, and basic fuzzy string matching is not very performative.

[17] "Uganda - Central Intelligence Agency." https://www.cia.gov/library/publications/the-world-factbook/geos/ug.html. Accessed 8 Jan. 2020.

A.4.1 Tradeoffs in Classification

The different spatial delineations speak to a tradeoff when it comes to prediction precision and weather precision. Predicting smaller delineations that are created based on climatic crop suitability (as agro-ecological zones and to a lesser extent, GAUL), would produce worse classification performance while creating more precise weather shocks. On the other hand, predicting based on ethnolinguistic boundaries (such as the GAUL layers) would lead to higher predictive power but less precision for weather shocks, since those regions contain multiple agro-ecological zones. Any averaging of weather indicators would contain data from multiple zones that have different climatic conditions. For instance, one climactic zone might be experiencing a drought, while the others in a region would not, and we would fail to pick up that weather event.

To use the ASAP data, however, it was necessary to use FAO GAUL administrative layers, which maps fairly well onto Ugandan agro-ecological zones. A comparison between GAUL and agro-ecological zones is presented in Figure 20.

To choose the correct level of spatial aggregation, I trained the XGBoost model on three levels: the GAUL layers (as in the main body of the paper), SAP Regions, which are for North, Central, East and Western regions of the country, and districts, of which there are 112. Below is the cross-validation tables for regions and districts.

Table 7: Results Cross-validation Folds of XGBoost Classifier Compared against a Random Classifier for Region

	Random Classifier	Calibrated XGBoost Classifier
CV Fold 1	0.302	0.884
CV Fold 2	0.313	0.886
CV Fold 3	0.311	0.912
CV Fold 4	0.303	0.895
CV Fold 5	0.307	0.912
CV Fold 6	0.303	0.891
CV Fold 7	0.309	0.906
CV Fold 8	0.309	0.895
CV Fold 9	0.300	0.894
CV Fold 10	0.310	0.909
CV Fold Mean	0.307	0.898

Note: Table shows results of stratified group cross-validation for a calibrated XGBoost classification model using SAP-Regions as a target. Dummy classifier that predicts from empirical distribution of target is shown for comparison.

Table 8: Results Cross-validation Folds of XGBoost Classifier Compared against a Random Classifier for Districts

	Random Classifier	Calibrated XGBoost Classifier
CV Fold 1	0.00000	0.00180
CV Fold 2	0.00000	0.00780
CV Fold 3	0.00000	0.00900
CV Fold 4	0.00000	0.00200
CV Fold 5	0.00000	0.00140
CV Fold 6	0.00000	0.01160
CV Fold 7	0.00000	0.00500
CV Fold 8	0.00000	0.00060
CV Fold 9	0.00000	0.00100
CV Fold 10	0.00000	0.00380
CV Fold Mean	0.00000	0.00440

Note: Table shows results of stratified group cross-validation for a calibrated XGBoost classification model using Districts as a target. Dummy classifier that predicts from empirical distribution of target is shown for comparison.

A.5 Classification Preliminaries and Details

I use two principle methods to classify drivers into different regions or agro-ecological zones: what we call a table classifier and a machine learning classifier. The table classifier creates a frequency table of surnames and the percentage of occurrences in the target variable that they belong to. For instance, if the surname "Akello" appeared in the North, 3/10 times of all appearances of "Akello", then that surname would be given a 30% chance of appearing in the North.

While the table classifier is straightforward, classification can only occur with names that appear in the nationally representative data set, or that appear in a form exactly like the one in the dataset.[18] The other method used was to train a machine learning classifier on the name data. In this case, we used a gradient boosting classifier with tf-idf weighted n-grams. Gradient boosting uses many regression trees to find the best "rule" of how to discriminate between one group and another.

N-grams are a method used in computational linguistics to break up a word into constituent parts for analysis. It is a technique ubiquitous across linguistics, machine learning, and natural language processing. For example, the name "Akello" as a 1-gram would be the one-letter variables of "A", "k", "e", "l", "l", "o"." As a 2-gram, we would have "Ak", "ke", "el", "ll", "lo", and then for 3-grams, 4-grams, etc… From then it is possible to count the number of occurrences of an n-gram in a particular word or to weight each gram in a specified way. One way is to use tf-idf (term-frequency, inverse document frequency) weighting of the

[18] Although fuzzy-matching was attempted with the table classifier, given that there is no matcher for any Ugandan language, the performance of this was too slow and often gave strange results.

n-grams to get a more informative set of variables to train on. Tf-idf weighting weights the frequency of a gram by how often that gram occurs in the whole dataset. These weight grams occur very often lower, and weights rarer grams more. This has the advantage of having the machine learning algorithm focus more on the part of the word that would give the most meaningful signal of coming from one label over another. The tf-idf transformed n-grams then become features or variables used for fitting a machine learning algorithm. In my case, we ran the model with 1,2,3 gram features. Using higher order n-grams did not yield better results, but increased computation time considerably.

The results of the training yielded promising results. We use three metrics to quantify the performance of each of the target variables: accuracy, precision, and F-score. Accuracy can be simply defined as the number of correct times the trained model was able to correctly identify the class of the target in the testing data.[19] If we simplify to a binary classification framework, where there were only two classes to classify into (cat or dog, for instance), we can formally write accuracy as:

$$Accuracy = \frac{TP}{TP + TN + FP + FN}$$

where TP is true-positive, TN is true negative, FP is false positive and FN is a false negative. This refers to how the model predicts the model in each class.

Precision and F-score were originally designed for binary classification models and are standard metrics for quantifying classification performance. But since there are four GAUL-SAP

[19]This means, for instance correctly identifying that the surname comes from the North, the South, the Center, or the East in terms of region.

regions and ten GAUL layers, precision and f-score are redefined to include multiple classes. For binary classification, precision is:

$$Precision = \frac{TP}{TP + FP}$$

Precision tells us the probability of correctly predicting the class, conditional on all the times it predicted an observation in that class. For multi-class classification, we calculate precision in a "one vs. all" fashion, where we treat the class of interest as one class and all other classes as another class. This reformulates the problem into C binary classification problems where C is the number of classes. Multi-class precision is then calculated as a weighted average of these C binary precision metrics.

The F1-score can be defined as the geometric mean of precision and recall, another metric used in classification.[20]

$$F1 = 2 \cdot \frac{Precision \cdot Recall}{Precision + Recall}$$

The F1-score is a better metric than accuracy as it is robust to false negatives and false positives.

[20] Recall can be defined as $TP/TP + FN$ and tells us how robust predictions are to false negatives. In this case, we omit the recall score, but they were comparable to precision.

A.6 Visualizing the Behavioral Changes of Drivers

After using the predictions to merge the Uber driver data and the weather shocks, it is now possible to visually explore whether there is a response by the drivers after weather shocks. Figure 21 presents each GAUL region's cumulative NDVI z-score with the average hours online of each driver matched to that GAUL region. Orange bands represent times when the Z-score dips below -1, a cutoff used for a severe event. Cutoffs for the plot and the regressions were used to be sufficiently low, while also having enough support to estimate meaningful coefficients. Although these graphs include seasonal effects, the figures show that hours seem to drop when NDVI is higher and vegetation is greener, and begin to increase when NDVI dips. For instance, in the South Eastern region (fourth row, first column), the very large dip in NDVI (which goes beyond the cutoff value of -1) around the end of 2016, coincides with an increase in hours. Figure 21 also shows that different GAUL regions are affected at different times, adding strength to the overall identification. Generally, however, the signal from these curves is noisy, which is what the switching regression framework is meant to account for.

A.6.1 Distributive Lags and Misclassification

Throughout the analysis in this paper, I will present an econometric specification that includes the lags of the shocks that I am interested in. These lag terms will seem to change by different magnitudes despite affecting one set of drivers. The explanation for this lies behind an implicit assumption that I have made to simplify the above examples: the number of regimes that are allowed to be treated within a given period. In reality, due to the

correlated nature of weather shocks, multiple groups of drivers can be treated and lags may be capturing different sets of drivers as treatment and control groups. For example, suppose that there is no misclassification and in period 1, regime 1 gets hit by a weather shock, but in period 2, groups 1 and 2 become treated. The ATE for period 1 will compare group 1 against groups 2 and 3:

$$ATT_{t=1} = E(Y_{i1}^1 - (Y_{i1}^2 + Y_{i2}^3))$$

In the second period, groups 1 and 2 will be compared against group 3:

$$ATT_{t=2} = E((Y_{i2}^1 + Y_{i2}^2) - Y_{i2}^3)$$

Without misclassification, there would be no issue, and the conditional independence assumption makes different treatment and control groups in this instance to be allowed. However, suppose that we have the same misclassification setup as in Equation 12. Then $ATT_{t=1}$ will become equivalent to $ATTR_1$ and $ATT_{t=2}$ will become $ATT_{t=2}^m$:

$$ATT_{t=1}^m = (\pi_{11} - \pi_{12})E(Y_{it}^1) - (\pi_{11} - \pi_{12})E(Y_{it}^2) - E(Y_{it}^3)$$

$$ATT_{t=2}^m = (\pi_{11} + \pi_{21})E(Y_{i2}^1) + (\pi_{12} + \pi_{22})E(Y_{i2}^2) - E(Y_{i2}^3)$$

The degree of bias in each average treatment effect will now depend on different misclassification probabilities.

A.7 Robustness of Results

A.7.1 Regression Results

In this section, we show the different sub-specification used for each regression that I ran. This involved running three specifications:

1. The "basic" Specification:

$$y_{it} = \sum_{t \in \{0 \ldots 8\}} \beta_t Shock_{r,t}$$

$$Female_i \gamma_1 + SharedVehicle_{it} \gamma_2 + \varepsilon_{irt}$$

This is the basic specification without any fixed effects, and with both demographic variables available.

2. Fixed Effects only

$$y_{it} = \sum_{t \in \{0 \ldots 8\}} \beta_t Shock_{r,t}$$

$$+SharedVehicle_{it} \gamma_1 + \alpha_i + \varepsilon_{irt}$$

And finally,

3. Twoway Fixed Effects

$$y_{it} = \sum_{t \in \{0 \ldots 8\}} \beta_t Shock_{r,t}$$

$$+SharedVehicle_{it} \gamma_1 + \alpha_i + \alpha_t + \varepsilon_{irt}$$

This was the specification used in the plots in the main body of the paper. The shock variable is either a continuous variable for the negative Z-score of Cumulative NDVI or a set of indicator variables for no shock, mild shock, and severe shock.

A.7.1.1 Severity Indicator for Hours Online

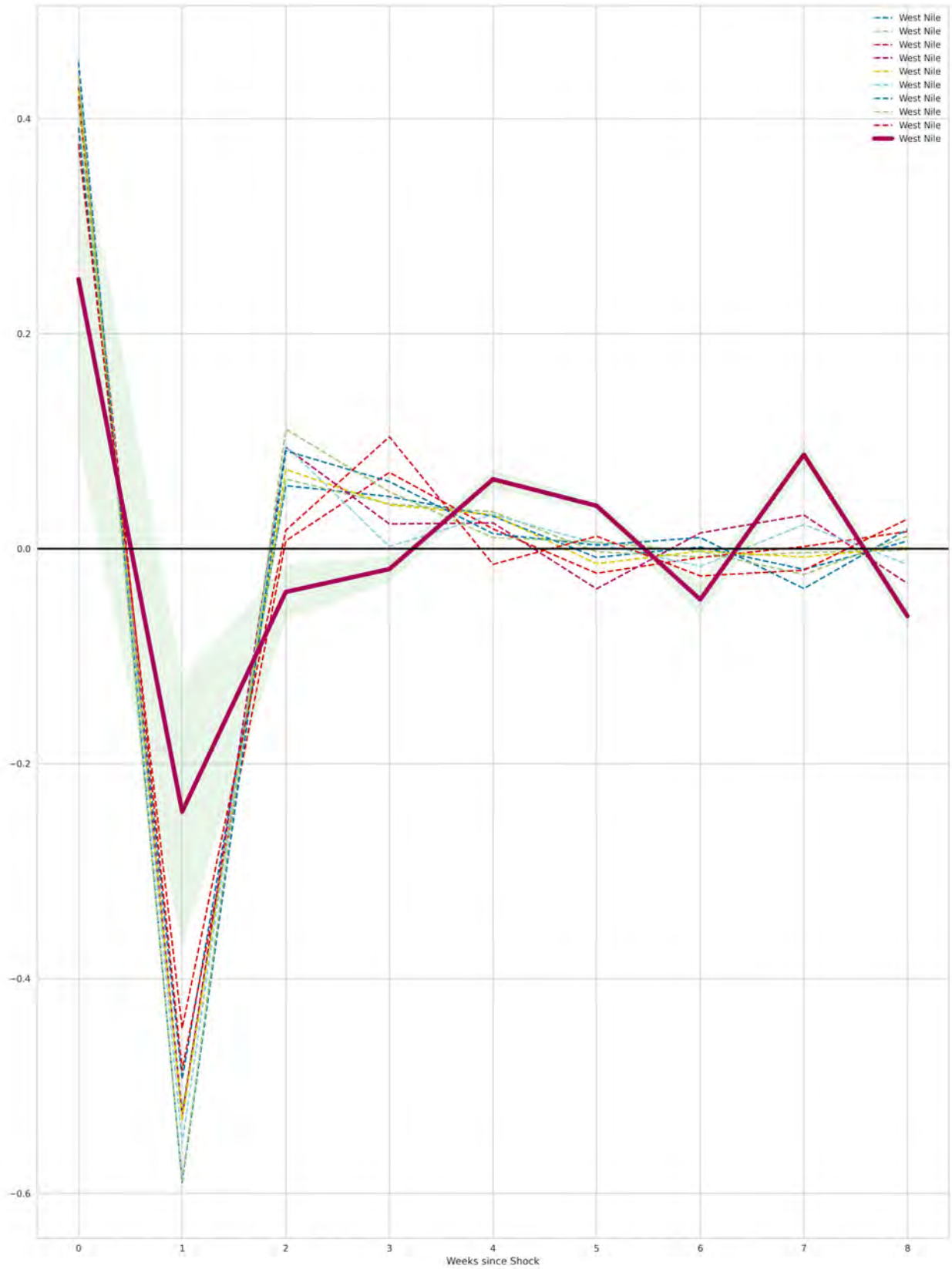

Figure 15: Extensive Margin Effects across GAUL Regions

Note: Coefficient plot shows the results of an interaction model of each of the distributed lags with each predicted GAUL region. Thick solid line represents effects for the Karamoja region. Shaded area denotes 95% confidence intervals. Standard errors clustered at the GAUL level. Total effects calculated with the delta method. N=104,117

Figure 16: Coping with Shocks

Notes: Source: LSMS-ISA Uganda 2016, 2018, 2019. Plot shows the % of households that reported relying on savings, unconditional help from friends and relatives or unconditional help from the government, as the primary way of coping with a flood or drought event within each GAUL region. Categories not shown include changing dietary patterns involuntarily, household members taking on more farm wage employment, household members migrating (was never the primary choice), obtaining credit (was never a primary choice), selling durable household assets, selling land/building, renting out land/building, distress sales of animal stocks, sending children to live elsewhere, reducing expenditures on health and education or other reasons.

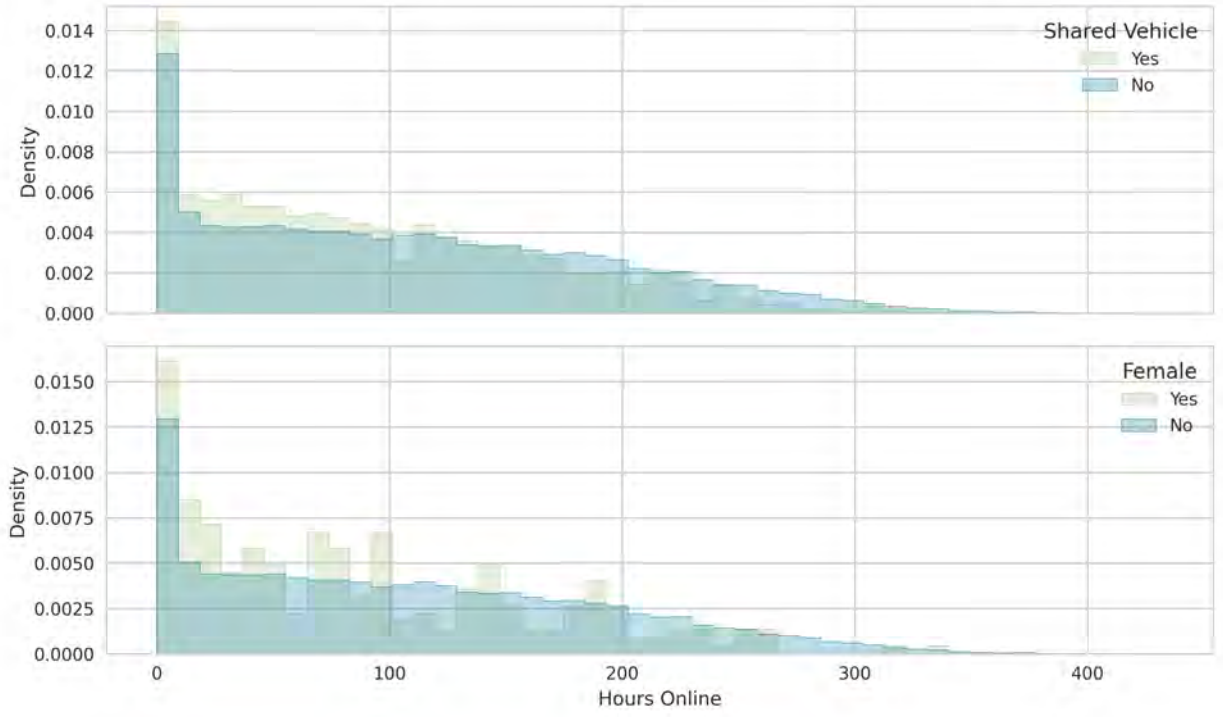

Figure 17: Histogram of Monthly Hours Online by Shared Vehicle Status and Gender

Note: Figure depicts weekly hours online. An active timeline for a driver is defined as all periods between the first and last time the driver was observed. Hours online trimmed at the 99th percentile. Shared Vehicle denotes those drivers that either share or rent out a vehicle for doing trips.

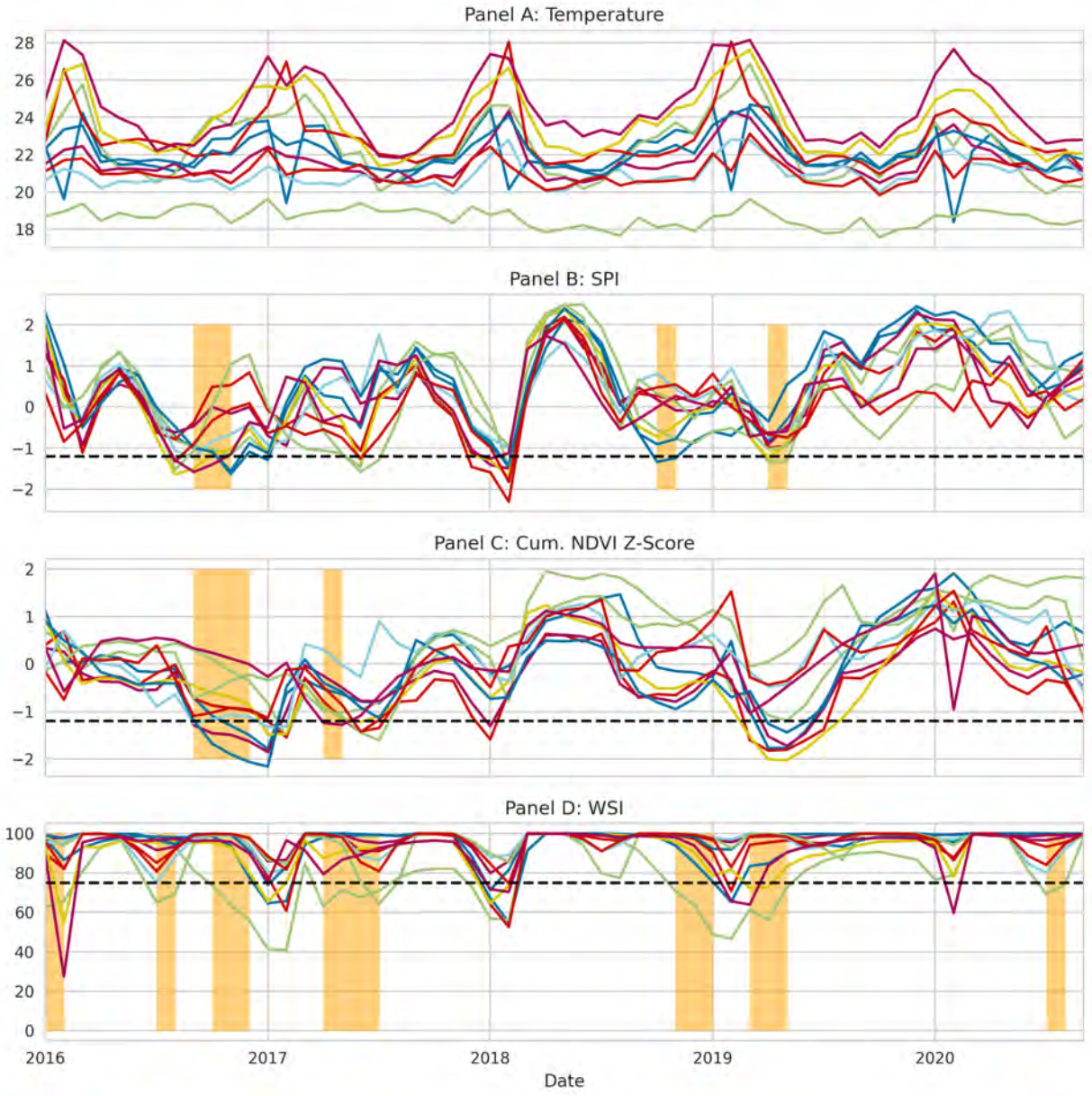

Figure 18: Weather Indicators with GAUL Regions

Note: Weather indicators downloaded from https://mars.jrc.ec.europa.eu/asap/, and grouped by GAUL regions. Shaded regions denote times of low biomass growth in one area and not in another, illustrating spatial variation. The dotted black line denotes an anomalous event, which is -1.2 for SPI and cNDVIz and 75 for WSI.

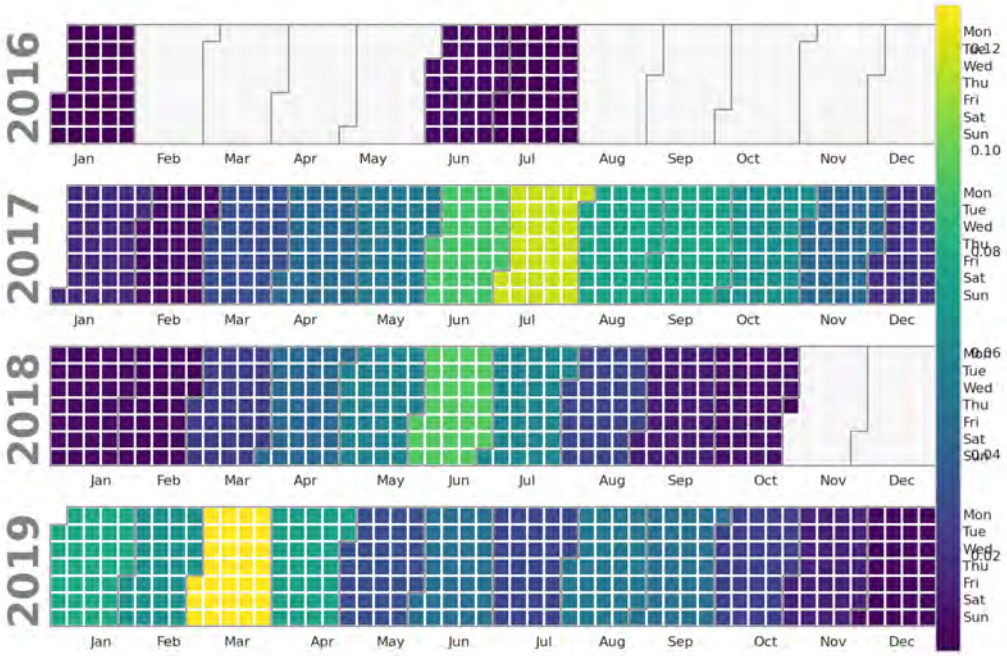

Figure 19: Percentage Reporting of Shocks due to Drought, Flood or Irregular Rains

Note: Data used was from LSMS-ISA 2015,2017,2019. The questionnaire asked whether a shock occurred, when it occurred and what effect it had on the household. Plot colormap normalized as percentage of all responses. Grey areas denote that month was mentioned at all. Percentages are raw.

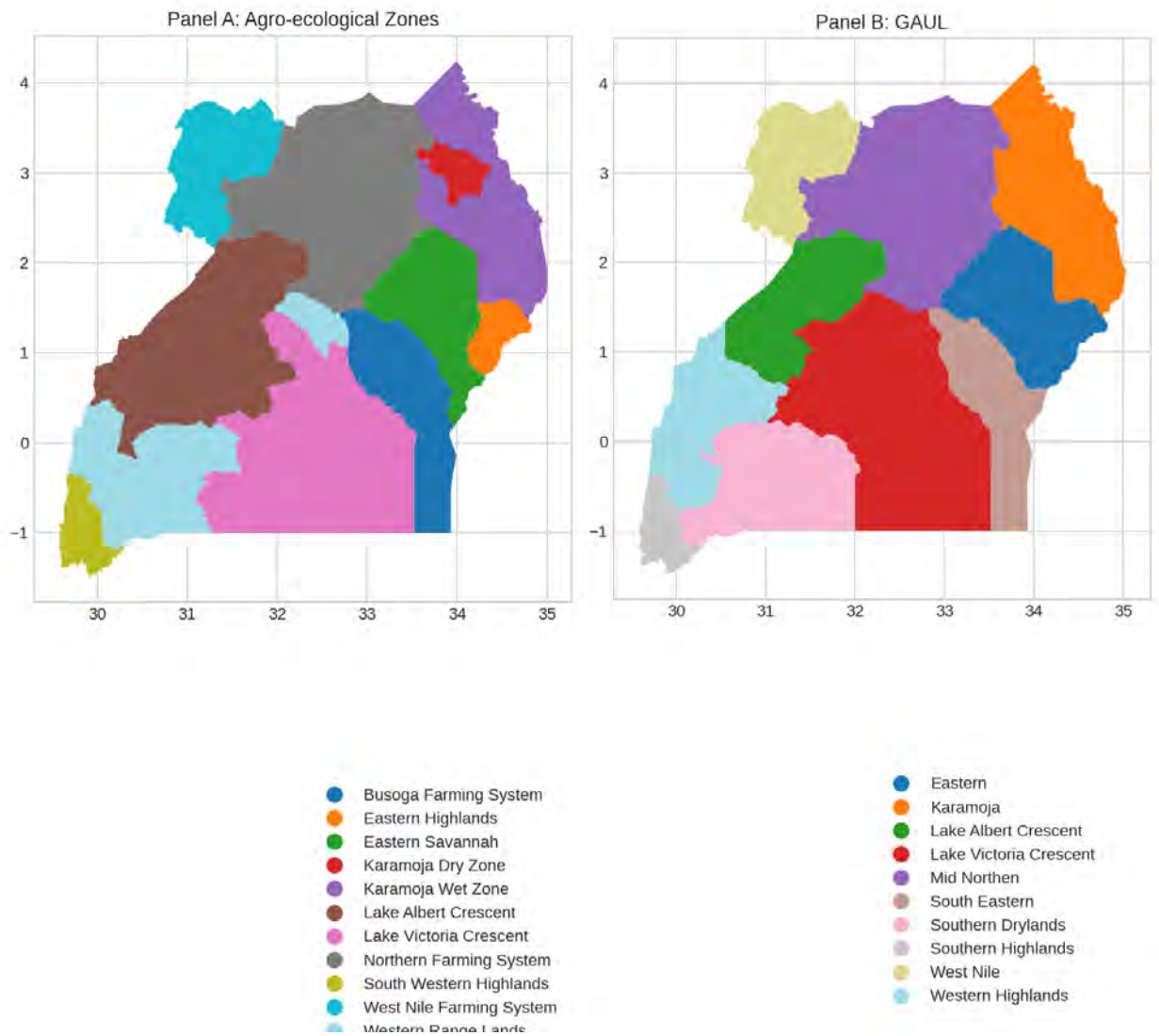

Figure 20: A Spatial Breakdown of GAUL and Agro-ecological Zones

Note: Figure depicts a comparison between GAUL Regions and Ugandan agro-ecological zones boundaries. GAUL boundaries determined by FAO. Agro-ecological zones determined by FAO.

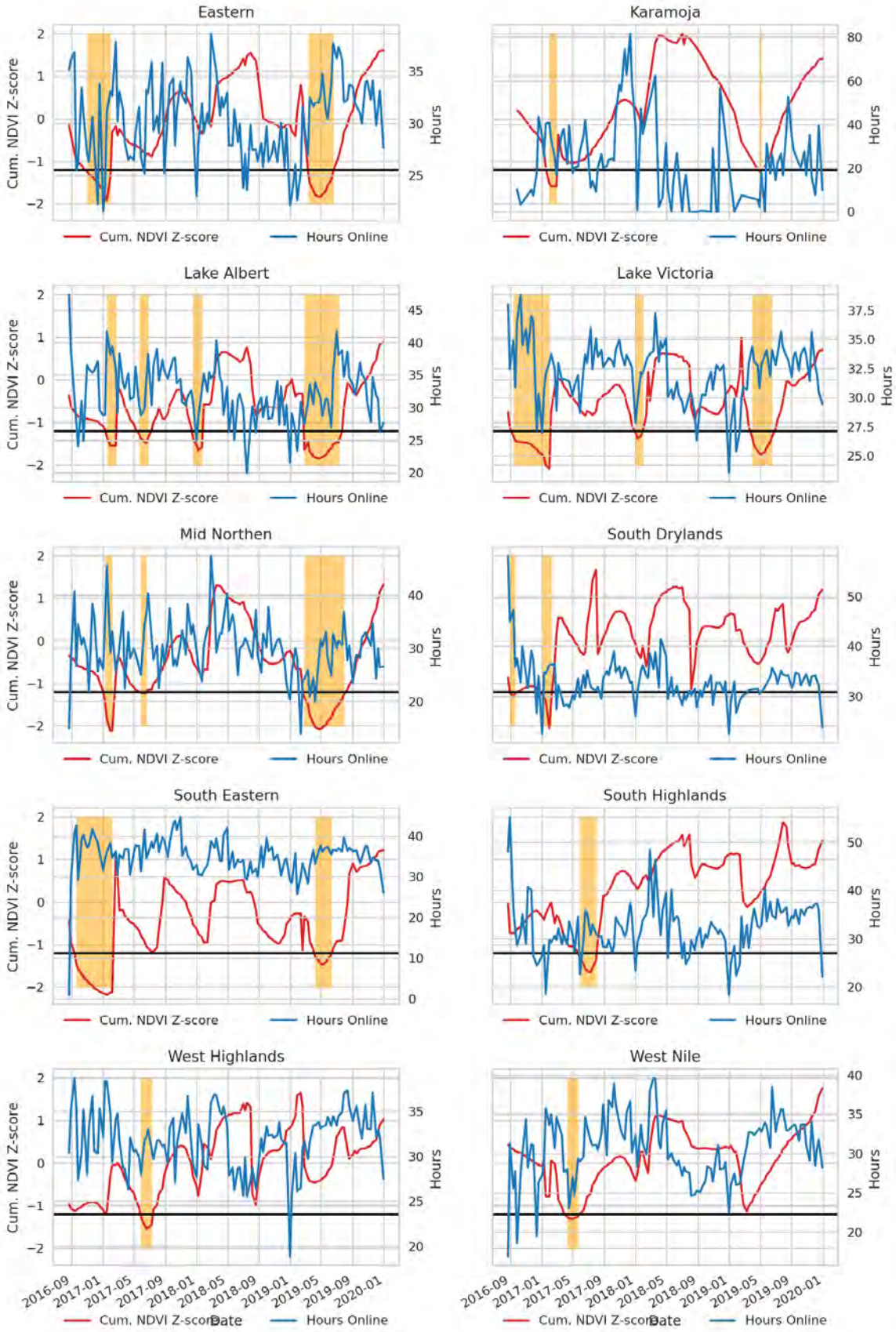

Figure 21: Hours Online and Weather Shocks

Note: Hours online resampled to the monthly level. Hours trimmed to the 99th percentile. Cumulative NDVI measure downloaded from https://mars.jrc.ec.europa.eu/asap/, and grouped by GAUL regions. Shaded

Table 9: Heterogeneous Response to Shocks

	OLS			Switching Regression		
	Hours Online	Hours Online	Hours Online	Hours Online	Hours Online	Hours Online
Mild Shock t	-0.117	0.511*	-0.266	-0.104	1.265**	3.684
	(0.336)	(0.288)	(0.200)	(0.779)	(0.394)	(3.126)
Severe Shock t	-0.134	1.233***	0.186	0.011	1.609*	5.515**
	(0.633)	(0.383)	(0.424)	(1.246)	(0.727)	(2.036)
Mild Shock t-1	-0.135	0.172	-0.321	-0.372	0.344	0.036
	(0.281)	(0.291)	(0.341)	(0.529)	(0.342)	(2.621)
Severe Shock t-1	0.682	0.896	0.559	0.792	0.986	1.355
	(0.607)	(0.609)	(0.384)	(1.134)	(0.866)	(2.177)
Mild Shock t-2	-0.222	0.006	-0.860**	-0.315	0.315	-0.372
	(0.439)	(0.462)	(0.410)	(0.719)	(0.575)	(1.300)
Severe Shock t-2	-0.399*	-0.282**	-0.918***	-0.404	-0.387	-3.545***
	(0.212)	(0.129)	(0.234)	(0.515)	(0.294)	(0.757)
Mild Shock t-3	0.125	0.470	0.033	0.192	1.137*	3.208***
	(0.447)	(0.311)	(0.558)	(0.697)	(0.571)	(0.947)
Severe Shock t-3	-0.451*	-0.227	-0.508**	-0.470*	-0.353	-0.253
	(0.243)	(0.232)	(0.257)	(0.251)	(0.234)	(0.481)
Mild Shock t-4	-0.641*	-0.432	-0.894	-0.992	-0.515	-0.935
	(0.348)	(0.446)	(0.556)	(0.651)	(0.509)	(1.170)
Severe Shock t-4	0.040	0.376	-0.221	0.250	0.632	0.866
	(0.343)	(0.322)	(0.265)	(0.764)	(0.608)	(0.846)
Mild Shock t-5	0.690	0.714*	0.400	0.519	1.054	1.618
	(0.503)	(0.376)	(0.451)	(0.633)	(0.682)	(0.955)
Severe Shock t-5	1.745***	1.870***	1.561***	2.058**	2.202**	2.730**
	(0.519)	(0.517)	(0.493)	(0.643)	(0.970)	(1.165)
Mild Shock t-6	-0.358**	-0.253	-0.449***	-0.617**	-0.075	-0.388
	(0.156)	(0.289)	(0.169)	(0.193)	(0.417)	(0.383)
Severe Shock t-6	-0.314*	-0.195	-0.541*	-0.032	0.023	-0.557
	(0.183)	(0.215)	(0.305)	(0.389)	(0.730)	(0.584)
Mild Shock t-7	-0.028	0.087	-0.338*	-0.144	0.385**	-0.285
	(0.253)	(0.104)	(0.176)	(0.415)	(0.147)	(0.298)
Severe Shock t-7	-0.407*	-0.115	-0.403	-0.137	0.131	-0.214
	(0.242)	(0.265)	(0.276)	(0.874)	(0.397)	(0.672)
Mild Shock t-8	0.668*	0.602**	-0.199	0.700	1.248**	0.476
	(0.384)	(0.304)	(0.349)	(0.494)	(0.428)	(0.713)
Severe Shock t-8	0.945**	1.178***	0.420	1.251	1.303***	1.806***
	(0.482)	(0.237)	(0.364)	(0.701)	(0.285)	(0.531)
Intercept	32.026***			31.890***		
	(0.505)			(0.478)		
Female	-6.174***			-6.175***		
	(1.652)			(1.740)		
Shared Vehicle	-2.644***	-6.219***	-6.090***	-2.631***	-6.221***	-6.158***
	(0.452)	(0.417)	(0.486)	(0.473)	(0.432)	(0.443)
Goodness of Fit	0.002	0.008	0.005	1104621.954	1049527.604	1047310.778
N	104117	104117	104117	104117	104117	104117
Driver FE	No	Yes	Yes	No	Yes	Yes
Time FE	No	No	Yes	No	No	Yes

Note: Standard errors clustered at the GAUL level. Analysis done with FAO GAUL boundaries disaggregated to 10 regions. "Time FE" refers to date fixed effects, "Driver FE" refers to driver fixed effects. drought intensity measured in terms of negative cumulative NDVI z-score over the growing period of a crop identified by ASAP. Unadjusted R^2 reported for Panel OLS, with Bayesian Information Criterion reported for switching regression models. No intercepts reported for fixed effects models. * p<0.10 ** p<0.05 ** p<0.01

A.7.1.2 Log Earnings with Continuous Shock Intensity

Table 10: Effect of Shock Intensity on Earnings

	OLS			Switching Regression		
	Log Earnings	Log Earnings	Log Earnings	Log Earnings	Log Earnings	Log Earnings
Shock Intensity t	0.020	0.074***	0.027***	0.040*	0.100***	0.248***
	(0.015)	(0.013)	(0.009)	(0.020)	(0.017)	(0.019)
Shock Intensity t-1	-0.036***	-0.029***	-0.035***	-0.062***	-0.053***	-0.127***
	(0.008)	(0.009)	(0.011)	(0.012)	(0.013)	(0.017)
Shock Intensity t-2	-0.003	0.015	-0.001	0.008	0.025	0.015
	(0.017)	(0.016)	(0.014)	(0.024)	(0.019)	(0.027)
Shock Intensity t-3	0.011	0.020	0.015	0.008	0.017	0.032
	(0.016)	(0.018)	(0.014)	(0.019)	(0.020)	(0.020)
Shock Intensity t-4	-0.010	-0.007	-0.011	-0.014	-0.010	-0.002
	(0.016)	(0.019)	(0.019)	(0.020)	(0.024)	(0.023)
Shock Intensity t-5	0.018	0.024	0.009	0.026	0.031	0.003
	(0.017)	(0.017)	(0.021)	(0.022)	(0.024)	(0.029)
Shock Intensity t-6	0.007	0.015	0.005	0.014	0.022	0.016
	(0.010)	(0.013)	(0.012)	(0.019)	(0.024)	(0.024)
Shock Intensity t-7	-0.002	0.011	-0.000	-0.015	0.007	-0.010
	(0.010)	(0.012)	(0.010)	(0.018)	(0.018)	(0.018)
Shock Intensity t-8	0.034***	0.019	0.006	0.051***	0.025*	0.043**
	(0.013)	(0.012)	(0.011)	(0.015)	(0.012)	(0.016)
Intercept	12.776***			12.773***		
	(0.017)			(0.017)		
Female	-0.350***			-0.350***		
	(0.078)			(0.082)		
Shared Vehicle	-0.000	-0.240***	-0.224***	0.000	-0.240***	-0.233***
	(0.025)	(0.012)	(0.016)	(0.026)	(0.013)	(0.013)
σ				1.143***	0.889***	0.881***
				(0.005)	(0.006)	(0.006)
Goodness of Fit	0.002	0.008	0.003	486473.803	434172.048	432125.668
N	104117	104117	104117	104117	104117	104117
Driver FE	No	Yes	Yes	No	Yes	Yes
Time FE	No	No	Yes	No	No	Yes

Note: Standard errors clustered at the GAUL level. Analysis done with FAO GAUL boundaries disaggregated to 10 regions. "Time FE" refers to date fixed effects, "Driver FE" refers to driver fixed effects. drought intensity measured in terms of negative cumulative NDVI z-score over the growing period of a crop identified by ASAP. Unadjusted R^2 reported for Panel OLS, with Bayesian Information Criterion reported for switching regression models. No intercepts reported for fixed effects models. * p<0.10 ** p<0.05 ** p<0.01

A.7.1.3 Severity Indicator for Log Earnings

Table 11: Heterogeneous Effects to Shock Intensity on Earnings

	OLS			Switching Regression		
	Log Earnings	Log Earnings	Log Earnings	Log Earnings	Log Earnings	Log Earnings
Mild Shock t	0.033	0.061***	0.009	0.055	0.116***	0.506**
	(0.027)	(0.019)	(0.015)	(0.039)	(0.033)	(0.176)
Severe Shock t	-0.000	0.057***	0.039*	0.000	0.079**	0.338***
	(0.025)	(0.012)	(0.021)	(0.042)	(0.027)	(0.091)
Mild Shock t-1	0.020	0.035**	0.007	0.019	0.049**	0.025
	(0.016)	(0.015)	(0.015)	(0.027)	(0.015)	(0.093)
Severe Shock t-1	0.024	0.024	0.009	0.022	0.010	-0.170
	(0.017)	(0.016)	(0.024)	(0.031)	(0.027)	(0.094)
Mild Shock t-2	-0.024	-0.006	-0.034**	-0.026	0.016	0.104*
	(0.017)	(0.017)	(0.013)	(0.026)	(0.018)	(0.051)
Severe Shock t-2	-0.046***	-0.039**	-0.069***	-0.055**	-0.047*	-0.099*
	(0.012)	(0.016)	(0.017)	(0.020)	(0.025)	(0.046)
Mild Shock t-3	0.011	0.028	0.004	0.021	0.057**	0.110*
	(0.018)	(0.021)	(0.021)	(0.030)	(0.025)	(0.051)
Severe Shock t-3	0.016	0.024	-0.000	0.041	0.045	0.041
	(0.017)	(0.018)	(0.016)	(0.025)	(0.026)	(0.037)
Mild Shock t-4	-0.010	-0.000	-0.002	-0.041	-0.007	0.008
	(0.023)	(0.026)	(0.018)	(0.032)	(0.039)	(0.064)
Severe Shock t-4	0.026	0.038	0.038	0.010	0.029	0.051
	(0.033)	(0.035)	(0.035)	(0.051)	(0.052)	(0.066)
Mild Shock t-5	-0.012	-0.015*	-0.021	-0.004	0.001	0.015
	(0.017)	(0.009)	(0.016)	(0.020)	(0.013)	(0.028)
Severe Shock t-5	0.004	0.007	-0.007	0.024	0.016	0.014
	(0.014)	(0.014)	(0.020)	(0.021)	(0.023)	(0.034)
Mild Shock t-6	0.004	0.013	-0.013	-0.008	0.025	-0.021
	(0.014)	(0.018)	(0.017)	(0.024)	(0.024)	(0.040)
Severe Shock t-6	-0.008	0.008	-0.010	-0.008	0.013	-0.015
	(0.016)	(0.013)	(0.019)	(0.024)	(0.018)	(0.026)
Mild Shock t-7	-0.007	-0.003	-0.025	-0.016	0.008	-0.011
	(0.020)	(0.015)	(0.022)	(0.031)	(0.025)	(0.033)
Severe Shock t-7	-0.002	0.014	-0.006	-0.012	0.010	-0.005
	(0.014)	(0.018)	(0.017)	(0.022)	(0.027)	(0.024)
Mild Shock t-8	0.023	0.014	-0.002	0.031	0.041	0.056
	(0.026)	(0.034)	(0.021)	(0.046)	(0.039)	(0.051)
Severe Shock t-8	0.047**	0.035	0.011	0.082**	0.049	0.075*
	(0.022)	(0.022)	(0.014)	(0.029)	(0.028)	(0.034)
Intercept	12.775***			12.770***		
	(0.016)			(0.015)		
Female	-0.347***			-0.347***		
	(0.079)			(0.082)		
Shared Vehicle	0.000	-0.240***	-0.224***	0.001	-0.240***	-0.234***
	(0.025)	(0.013)	(0.016)	(0.027)	(0.013)	(0.014)
Goodness of Fit	0.001	0.004	0.003	486648.029	434632.843	432390.754
N	104117	104117	104117	104117	104117	104117
Driver FE	No	Yes	Yes	No	Yes	Yes
Time FE	No	No	Yes	No	No	Yes

Note: Standard errors clustered at the GAUL level. Analysis done with FAO GAUL boundaries disaggregated to 10 regions. "Time FE" refers to date fixed effects, "Driver FE" refers to driver fixed effects. drought intensity measured in terms of negative cumulative NDVI z-score over the growing period of a crop identified by ASAP. Unadjusted R^2 reported for Panel OLS, with Bayesian Information Criterion reported for switching regression models. No intercepts reported for fixed effects models. * p<0.10 ** p<0.05 ** p<0.01

A.7.2 Extensive Margin with Continuous Shock Intensity

Table 12: Probability of Working as a Result of Shock Intensity

	OLS			Switching Regression		
	Hours Online > 0	Hours Online > 0	Hours Online > 0	Hours Online > 0	Hours Online > 0	Hours Online > 0
Shock Intensity t	0.031***	0.032***	0.034***	0.090***	0.082***	0.419***
	(0.003)	(0.003)	(0.003)	(0.013)	(0.011)	(0.015)
Shock Intensity t-1	-0.039***	-0.038***	-0.046***	-0.124***	-0.109***	-0.524***
	(0.003)	(0.003)	(0.006)	(0.018)	(0.014)	(0.025)
Shock Intensity t-2	0.008	0.009**	0.008	0.036***	0.032***	0.069***
	(0.005)	(0.005)	(0.006)	(0.009)	(0.008)	(0.018)
Shock Intensity t-3	-0.002	0.001	-0.000	-0.009	-0.004	0.041***
	(0.005)	(0.005)	(0.006)	(0.007)	(0.006)	(0.010)
Shock Intensity t-4	0.008*	0.007*	0.006	0.017***	0.015***	0.027**
	(0.004)	(0.004)	(0.004)	(0.004)	(0.004)	(0.008)
Shock Intensity t-5	-0.001	-0.001	-0.002	-0.003	-0.002	-0.006
	(0.006)	(0.005)	(0.005)	(0.005)	(0.005)	(0.007)
Shock Intensity t-6	-0.003	-0.004	-0.006*	-0.004	-0.005	-0.005
	(0.004)	(0.004)	(0.003)	(0.004)	(0.004)	(0.003)
Shock Intensity t-7	0.002	0.003	0.002	0.002	0.004	-0.004
	(0.004)	(0.004)	(0.005)	(0.005)	(0.005)	(0.007)
Shock Intensity t-8	-0.000	-0.000	-0.002	-0.000	-0.001	0.001
	(0.002)	(0.002)	(0.003)	(0.003)	(0.003)	(0.007)
Intercept	0.932***			0.932***		
	(0.001)			(0.002)		
Female	-0.007			-0.006		
	(0.009)			(0.010)		
Shared Vehicle	-0.058***	-0.059***	-0.055***	-0.057***	-0.058***	-0.049***
	(0.002)	(0.004)	(0.004)	(0.003)	(0.004)	(0.004)
σ				0.256***	0.238***	0.224***
				(0.002)	(0.002)	(0.003)
Goodness of Fit	0.005	0.004	0.004	175914.474	160652.984	156162.123
N	104117	104117	104117	104117	104117	104117
Driver FE	No	Yes	Yes	No	Yes	Yes
Time FE	No	No	Yes	No	No	Yes

Note: Standard errors clustered at the GAUL level. Analysis done with FAO GAUL boundaries disaggregated to 10 regions. "Time FE" refers to date fixed effects, "Driver FE" refers to driver fixed effects. drought intensity measured in terms of negative cumulative NDVI z-score over the growing period of a crop identified by ASAP. Unadjusted R^2 reported for Panel OLS, with Bayesian Information Criterion reported for switching regression models. No intercepts reported for fixed effects models. * p<0.10 ** p<0.05 ** p<0.01

A.7.3 Pre-trends

In this section, we show the pre-trends regressions for each specification used in the paper. These plots will only include the two-way fixed effects specification for each model.

A.7.3.1 Hours Online with Continuous Shock Intensity

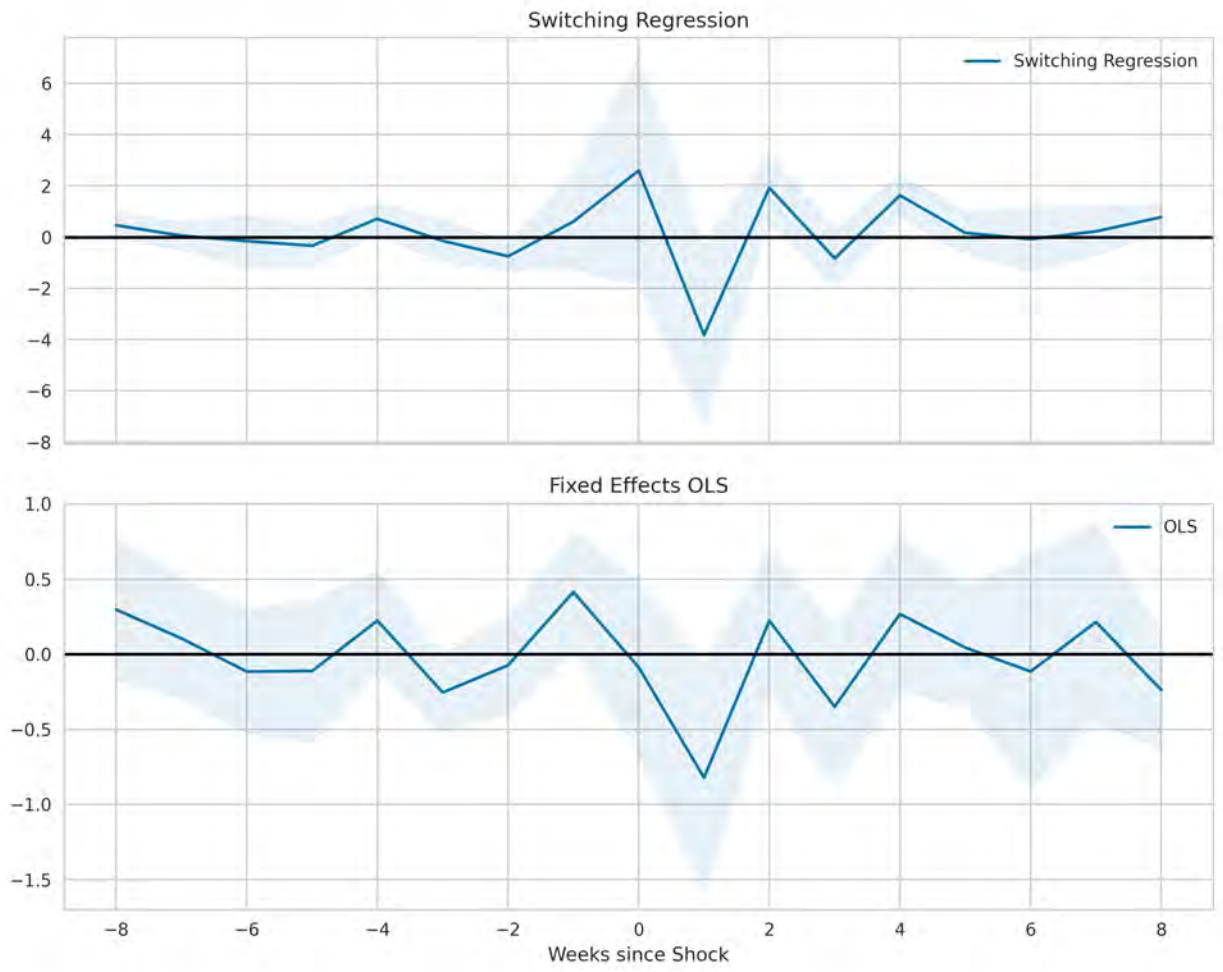

Figure 22: Pre-trend for Hours Online

Note: Figure shows pre-trend for hours online using 8 week lags and leads. Shaded areas show 90% and 95% confidence intervals. Standard errors clustered at the GAUL level.

A.7.3.2 Severity Indicator for Hours Online

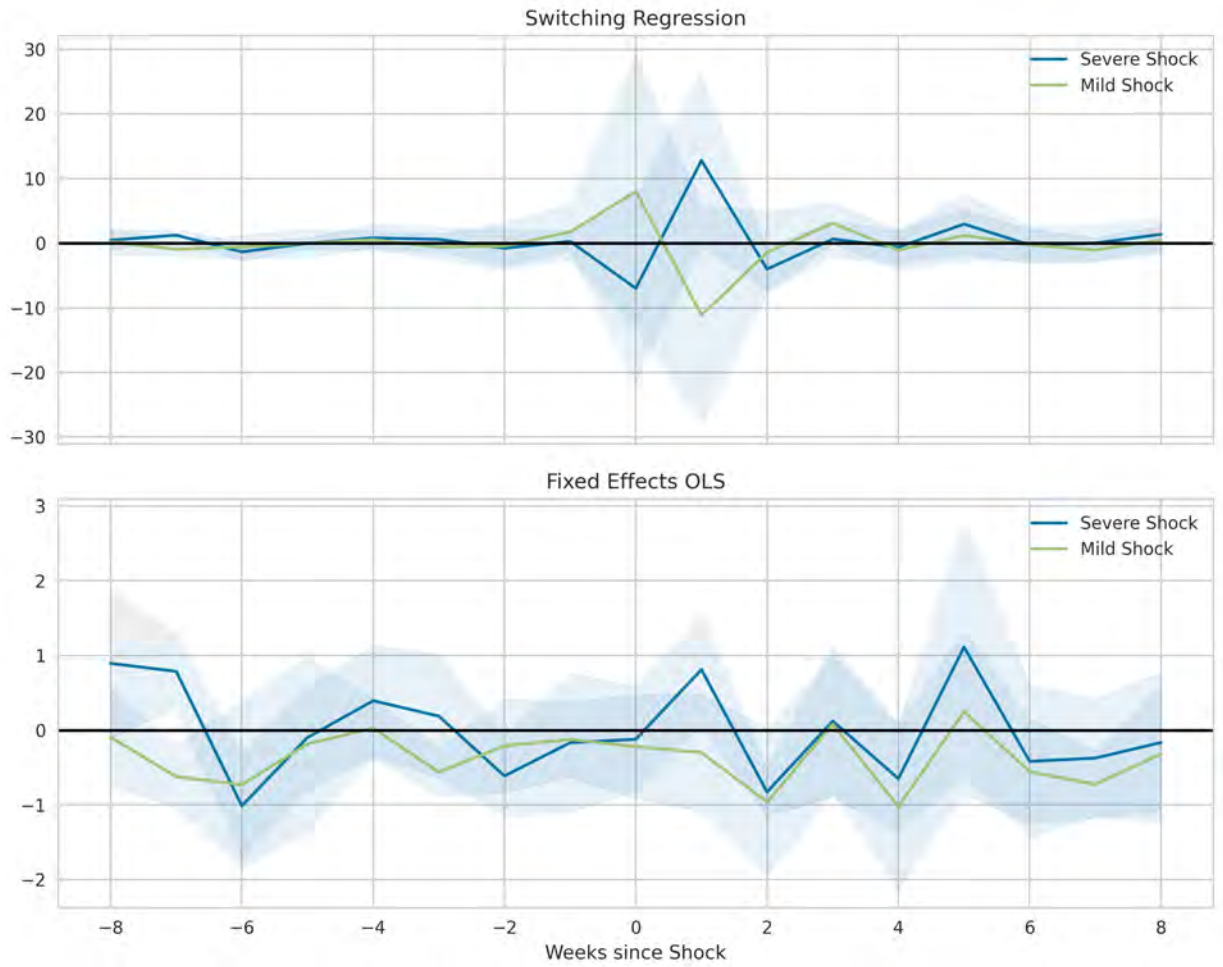

Figure 23: Pre-trend for Heterogeneous Effects to Shock Intensity on Hours Online

Note: Figure shows pre-trend for hours online using 8 week lags and leads. Independent variable was an indicator for mild drought (-1.2 < cNDVIz < -0.9) and severe drought (cNDVIz < -1.2). Shaded areas show 90% and 95% confidence intervals. Standard errors clustered at the GAUL level.

A.7.3.3 Log Earnings with Continuous Shock Intensity

A.7.3.4 Severity Indicator for Log Earnings

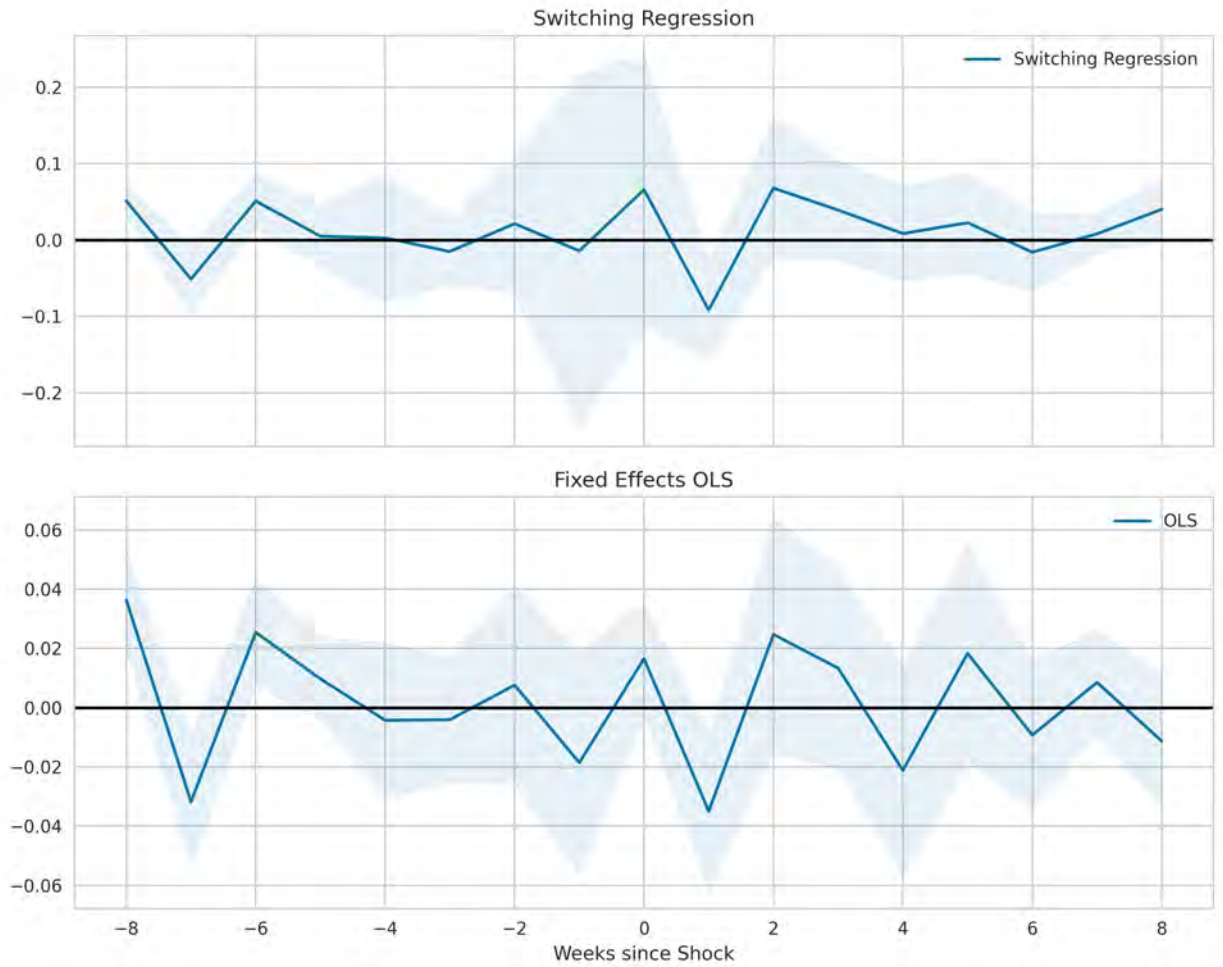

Figure 24: Pre-Trend for Effect of Shock Intensity on Log Earnings

Note: Figure shows pre-trend for log earnings using 8 week lags and leads. Shaded areas show 90% and 95% confidence intervals. Standard errors clustered at the GAUL level.

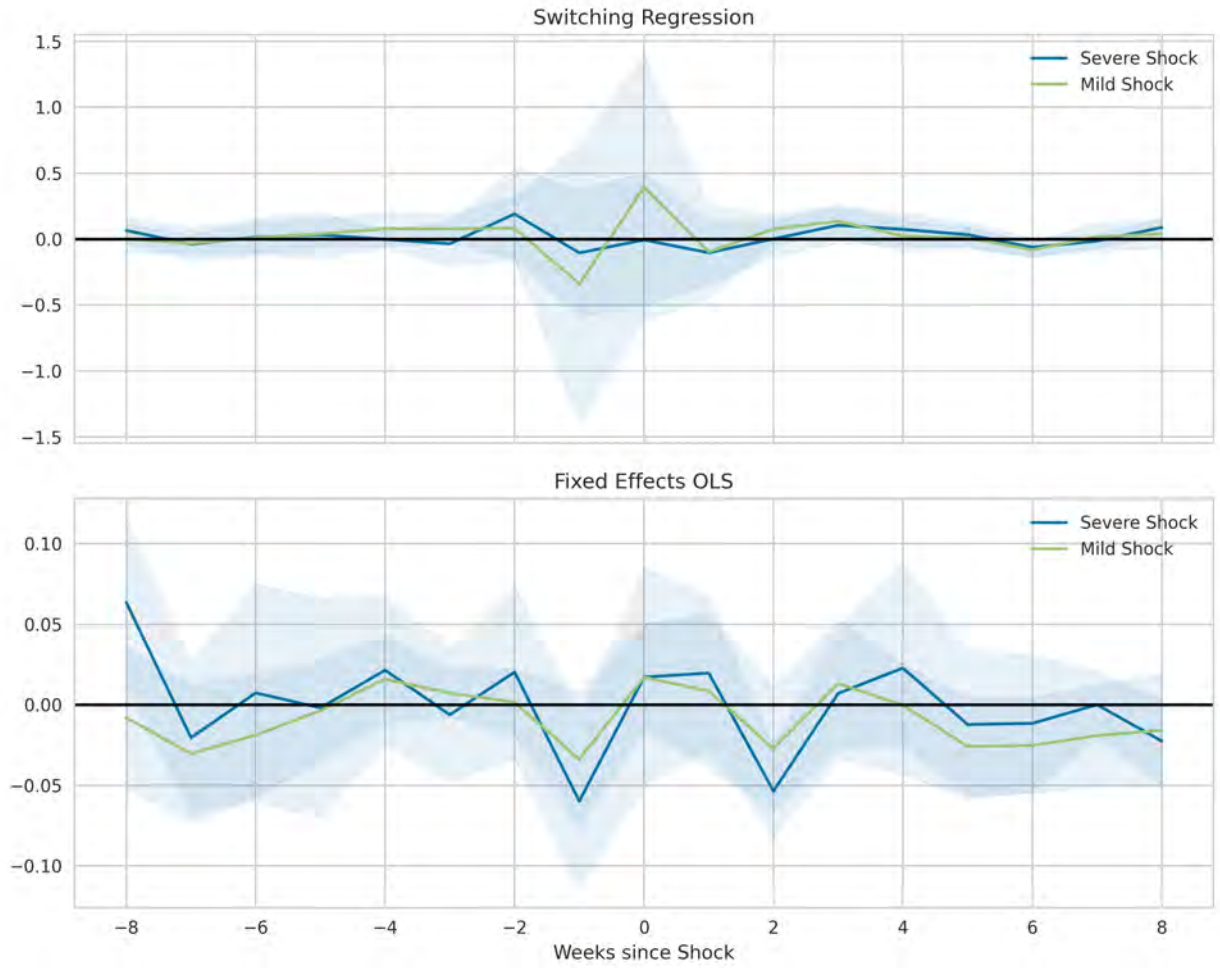

Figure 25: Pre-trend for Heterogeneous Effects of Shock Intensity on Log Earnings

Note: Figure shows pre-trend for log earnings using 8 week lags and leads. Independent variable was an indicator for mild drought (-1.2 < cNDVIz < -0.9) and severe drought (cNDVIz < -1.2). Shaded areas show 90% and 95% confidence intervals. Standard errors clustered at the GAUL level.